

The Tagin Annotation Rule Book

The standard for writing, normalizing, segmenting and annotating the Tagin language of Arunachal Pradesh: orthography, tokenization, part-of-speech tags and named entities.

Edition	5.0 · 29 September 2026
Language	Tagin · ISO 639-3 <code>tgj</code> · Glottolog <code>tagi1241</code>
Orthography	Modified Roman Script for Tagin (MRST)
Tagsets	13 Universal POS tags · 5 named-entity types (BIO)
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Corrections. Corrections, counter-examples and proposals are welcome from speakers, teachers, scholars and annotators. Chapter 22 describes how a proposal becomes a rule. When native speakers and community reviewers disagree with this book, the book is corrected.

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Preface

Tagin is spoken by about sixty thousand people in the hills of the Upper Subansiri and neighbouring districts of Arunachal Pradesh. It carries the memory of the Tagin people: their ancestors and spirits, their rituals and festivals, their villages and rivers, and the everyday speech of homes and markets. Like many languages of the region, it has been written in several ways, by different people, for different purposes, and until now there has been no shared agreement on how Tagin text should be spelled, divided into words, and labelled for use in dictionaries, school materials, archives and language technology.

This book is that agreement. It sets out, in one place, how to

- **write** Tagin in a single standard Roman spelling, and convert older spellings into it (Part II);
- **segment** Tagin text into tokens and sentences (Part III);
- **tag** every token with its part of speech (Part IV);
- **mark** the names of people, places, organisations, dates and culturally significant beings and practices (Part V);
- **create, check and release** new annotated data to a known standard of quality (Part VI); and
- **care** for Tagin knowledge in a way the community can trust (Part VII).

Scope

The rules in this book apply to **any Tagin text**: transcribed speech and oral narratives, chants and ritual speech, school readers, dictionaries, Bible translations, newspapers, and social-media writing. They are independent of any particular dataset or tool. Wherever a dataset, a program or a project follows this book, it should say which edition it follows (Chapter 20).

The book does not try to be a full grammar or dictionary of Tagin. It contains exactly as much grammar as an annotator needs to make a correct decision, and it points to the grammars, dictionaries and studies on which it rests (Appendix K).

How the rules were made

Every rule in this book can be traced to evidence: the Tagin dictionary and its grammar chapters, parallel Tagin–English texts, the usage of fluent speakers, the decisions recorded in the decision register (Appendix G), and published scholarship on Tagin and the Tani languages. Where Tagin allows more than one form, the book describes the variation and keeps it. Where the evidence was once unclear, the open question was taken to speakers and settled, and the ruling is recorded with its reasons.

The tagsets follow international practice, so that Tagin resources can stand beside those of other languages: parts of speech use the Universal Dependencies tagset, and named entities use the widely adopted BIO labelling scheme, extended with a **CULT** type for the beings, rituals, roles and objects that give Tagin culture its shape.

What is new in edition 5.0

This edition completes a native speaker’s review of the whole Tagin New Testament in the Mooya spelling. Particles are now written apart even in forms that earlier editions listed as single words: *lvga bv* “because”, *bunu gv* “their”, *hugu lo* “where”, *gvri la* and *ngo gv* “my”. The words that stay whole, and the pairs that are different words when joined and when split (*soka / so ka*), are now listed. Paired compounds, conversions from the Mooya spelling and loanword spellings gain new examples. Every ruling is recorded in the decision register (R41–R44). Because earlier rules are reversed, this is a major edition: data prepared under an earlier edition **SHOULD** be updated before it is combined with data prepared under this one (§22.1).

Edition 4.0, of the same date, made “you all” *nunu* and “you two” *nunyi*, wrote a name and *nyi* as two tokens (*jisu nyi*), wrote the future *re* and the command *to* apart from the verb (*vngji re*, *vngro to*) and “oneself” as *atu boongv* (R33–R40).

Acknowledgements

This book rests on the knowledge of Tagin speakers, elders, priests, teachers and writers, and on the work of the lexicographers, translators and linguists who documented the language before us. Their contribution is gratefully acknowledged. Any errors are the maintainer’s, and corrections are welcome.

Tungon Dugi
29 September 2026

How to use this book

If you want to...	Read
understand the language before annotating	Part I (§1–§3)
type or convert Tagin text into the standard spelling	Part II (§4–§6)
split text into tokens and sentences	Part III (§7–§8)
assign part-of-speech tags	Part IV (§9–§11)
assign named-entity tags	Part V (§12–§16)
run an annotation project, train annotators, measure quality	Part VI (§17–§22)
handle sacred, personal or community-owned material	Part VII (§23)
look up a word, marker, name, number or month	Appendices A–E
see complete annotated sentences	Appendix F
check a ruling or its reasons	Appendix G
have the essentials on one page	Appendix H

Guiding principles

1. **The community has the final say.** When native speakers, the Tagin Cultural Society or community reviewers disagree with this book, the book is corrected (Chapter 22).
2. **Describe the language; don’t “correct” it.** Dialect and register variants are kept (§6.2). Only spelling *inconsistencies* are normalized.
3. **Tag what the word means in its sentence.** The same written form can be a particle, a pronoun or a noun, and the same noun can be an ordinary word or the name of a deity. Meaning in context decides, never the form alone.
4. **Every decision can be traced to its evidence.** Each rule rests on the Tagin dictionary and grammar, parallel translations, speakers’ judgements, recorded decisions (Appendix G) or published scholarship (Appendix K).
5. **One rule, applied everywhere.** An adopted rule is applied to every text and to its translation at the same time, and is recorded in the edition history (Appendix L).
6. **When you don’t know, don’t guess.** Mark the item as uncertain and send it to adjudication (§18.4) rather than inventing a tag.

Conventions used in this book

Convention	Meaning
kaanam	A Tagin form in the standard spelling (MRST, Chapter 4), or a tag, label or literal character.

Convention	Meaning
<i>Kaanam</i>	A form quoted <i>as written in another source</i> , usually the Tagin dictionary, in that source's own spelling.
“see”	An English gloss or translation.
takar/PROPN/B-PER	A token written as <i>word/POS/NER</i> , used in examples.
§2.1	A cross-reference to a chapter or section of this book.
MUST, MUST NOT	An absolute requirement: data that breaks it does not conform to this standard.
SHOULD, SHOULD NOT	A strong recommendation: depart from it only for a documented reason.
MAY	Something that is allowed but optional.

The words **MUST**, **SHOULD** and **MAY** are used in the sense of the Internet standards BCP 14 (RFC 2119 and RFC 8174), and only when written in capitals as above.

The Tagin names used in examples (takar, tapa, yaro, tara and others) are common Tagin personal names chosen as illustrations. They do not refer to real individuals.

PART I

The Language

CHAPTER 1

Tagin at a glance

Property	Value
Family	Sino-Tibetan → Tani → Western Tani → Subansiri → Bangni-Tagin
Codes	ISO 639-3 <i>tgj</i> ; Glottolog <i>tagi1241</i>
Speakers	about 62,900 (Census of India 2011)
Homeland	Upper Subansiri district, Arunachal Pradesh (Daporijo, Dumporijo, Taliha, Siyum, Nacho, Giba, Limeking...), and neighbouring parts of West Siang, Papum Pare and Kra Daadi. About 500 speakers live in the adjoining Tibet Autonomous Region.
Status	<i>Vulnerable</i> (UNESCO Atlas of the World's Languages in Danger)
Close relatives	Bangni and Na (the same Bangni-Tagin subgroup); Nyishi; Galo, Adi, Apatani and Mising (the wider Tani group)
Mra	Mra (about 350 speakers) is administratively counted as “Tagin” but may be a distinct language. Keep Mra material in separate, clearly labelled collections.
Typology	Verb-final (subject–object–verb). Agglutinating: case, tense, aspect and mood are shown by particles that follow the word. Roots have one or two syllables.
Writing	Several Roman-based spellings are in use (Chapter 5). This book standardises on the Modified Roman Script for Tagin (MRST) (Chapter 4).

1.1 Registers

Tagin has an everyday register and a **ritual register** used in chants and priestly speech (the speech of the *nyibu*). The ritual register has its own vocabulary: the sun is *donyi* in everyday speech but *vjwng* in ritual speech (§D.2).

Rule

Ritual words are kept exactly as they are. They **MUST NOT** be “normalized” to their everyday equivalents.

1.2 Regions and varieties

The Tagin grammar distinguishes the speech of “upper” and “lower” Tagin speakers, and speakers vary in a number of regular ways (§2.3). All of these varieties are equally Tagin. Annotation records them as they are and gives equivalent forms the same tags.

CHAPTER 2

The sound system

2.1 Vowels

Tagin has **seven vowels**: a, e, i, o, u, v and w. **Each can be short or long, and length changes meaning.**

Standard	Dictionary spelling	IPA	Short example	Long example
a	a	a	abu father, patu half	aanam to come, kaanam to see
e	e	e	kedv land, selu today, re future marker	bee chant, meeru yesterday, seenv tree
i	i	i	j inam to give, nyi object marker	j iinam to plant crops, nyii person
o	o	o	oppo alcohol, oku leaf	oo vegetable, doogum thunder
u	u	u	usum urine, agu hot	uum hole
v	e	ə	v lw stone (dict. <i>elv</i>), lvga for (dict. <i>lega</i>), gv of (dict. <i>ge</i>)	vvro go away
w	v	ɪ	lvbw knee (dict. <i>levv</i>), hwdw when (dict. <i>hedv</i>)	bww he, she

The Tagin dictionary writes both e [e] and v [ə] as *e*. The standard keeps them apart: *seene* “tree” is seenv, with both vowels, and *lega* “for” is lvga (§5.1).

Minimal pairs that annotators must keep apart

j inam (give) vs j iinam (plant) · nyi (object marker for people) vs nyii (person) · nam (verbal-noun ending; “you”) vs naam (house).

2.2 Consonants

Group	Standard letters	Notes
Labials	b, p, m	There is no /v/ sound. English <i>v</i> becomes b, and English <i>f</i> becomes p.
Velars	k, g, ng [ŋ]	No aspirated <i>kh</i> or <i>gh</i> in native words.
Palatals	ch [tʃ], j [dʒ], ny [ɲ]	These never end a word. The dictionary writes <i>c</i> where the standard writes ch.
Dentals	t, d, n	t and d rarely end a word; n often does.
Fricative	s	

2. The sound system

Group	Standard letters	Notes
Glottal	h	only at the start or in the middle of a word
Lateral	l	
Flap	r	
Glide	y	never ends a word

Tagin has lexical tone, but tone is **not written** in the standard orthography (§5.3).

2.3 Regular variation between speakers

The following alternations are **not errors**. Keep both forms (§6.2) and give them the **same tags**.

Alternation	Examples	Who
m ~ b	mino ~ bindo (says); mw ~ bw (himself)	some speaker groups
l ~ n	lasuk ~ nasuk (wash hands); laanam ~ naanam (take); bulu ~ bunu (they); ngulu ~ ngunu (we); mwlww ~ mwnww (everyone)	some speaker groups
d ~ j at the start of a syllable	adin ~ ajin (meat); dibe ~ jibe (range); dvvnam ~ jiinam (plant)	some words
inserted ng	setu ~ sengtu (stump); yalu ~ yalung (quarrelsome); kobu ~ kobung (mouse); kamduu ~ kamdung (lazy)	more common in the upper region
h ~ y in question words	<i>hugu</i> ~ <i>yogu</i> (what); <i>hedv</i> ~ <i>yedv</i> (when) (dictionary spellings)	lower region
initial h dropped	hvbv ~ vbv, hvjak ~ vjak, hugu ~ ogo	the “Mooya” written variety (§5.2)
final -ng ~ long vowel	vvming ~ vvmii; chvbwng ~ chvbww	across speakers
e ~ i, o ~ u, a ~ o	amen ~ amin (name); aakur ~ aakor; mintam ~ mintom (tell)	across speakers
initial a dropped	alak ~ laak (hand); anyik ~ nyik (eye); agam ~ gaam (word, language)	some speakers

CHAPTER 3

A grammar sketch for annotators

This chapter gives enough grammar to tag correctly. It follows the grammar chapters of the Tagin dictionary (Appendix K). Dictionary examples are quoted *in italics, in the dictionary's own spelling*; examples from annotated text are in code, in the standard spelling.

3.1 Words and word order

- **Roots have one or two syllables.** A syllable can be a root on its own (kaa see, chin know, ma not, du present tense, cho uncertainty), and roots combine: kaachin “know by seeing”, kaama “not seen”, kaadu “seeing”, chinma “don’t know”.
- **Word order is subject – object – verb**, followed by the particles: ngo hookv dapo chaare . “I’ll go to Dapo later.”
- **Particles follow the word they belong to.** In the standard they are written as **separate words** (§7.2).

3.2 Case markers

Case markers are tagged **PART**. The exceptions are the place words so, ho and hoka, which are demonstratives (DET), and soka, which is an adverb (ADV). Examples are quoted verbatim from the Tagin dictionary, in its own spelling (*e* = standard *v*, *v* = standard *w*; §5.1).

Function	Standard marker	Dictionary	Example (dictionary spelling)
Subject (nominative, optional)	v	e	<i>Nyii e jokdu.</i> “The men are running.”
Object / recipient, non-human	vm, am	a / am / em	<i>Tape am kvvto.</i> “Boil the pumpkin.” · <i>Ikki am adin jito.</i> “Give meat to the dog.”
Object / recipient, human	nyi	nyi	<i>Yaro nyi isi jito.</i> “Give water to Yaro .” · <i>nyii mwlww takar nyi kaagia do .</i> “Everyone likes Takar .”
Location, here / now	so, soka (fixed spot)	so / soka	<i>No so aato.</i> “You come here.” · <i>Dapo soka doto.</i> “Stay in Dapo.”
Location, general	lo	lo	<i>Naam lo doto.</i> “Stay at home.”
Location, there / then (far)	ho, hoka	ho / hoka	<i>Abu no Dapo ho doto.</i> “Father, you stay at Dapo.”
Direction + location (down / up / level)	bvlo / tvlo / vlo	belo / telo / elo	<i>Seene ao telo geyca to.</i> “Climb to the top of the tree.”

3. A grammar sketch for annotators

Function	Standard marker	Dictionary	Example (dictionary spelling)
With / from / by means of	lok v	loke	<i>Tapar Yagv loke seene teedo.</i> “Tapar cuts the tree with an axe.” · <i>Ngo Ajen si Dapo loke.</i> “My friend is from Dapo.”
Towards / for	bv	be	<i>Abu lega be naam meto.</i> “Build a house for father.”
Possession (“of, ’s”)	gv, kv	ge / ke	<i>Talo ge naam e hugu lo?</i> “Where is Talo’s house?” · <i>So Oryok si noke.</i> “This dao is yours.”
Plural	v d w (also v d w v, v d w a, g w d w, w d w)	edv	kristan v d w gv “of the Christians”

Rule

nyi marks a **person** who is the object or recipient of the action. It is never the agent. The noun “person, people” is a different word, nyii (§11.1).

3.3 Pronouns

Person	Singular	Dual	Plural
1st	ngo (I); object ngam / ngom / nga (me); possessive ngo gv / ngokv	ngunyi (we two)	ngunu / ngu lu / ngun / ngul (we)
2nd	no (you); object nam / nom (you); possessive no gv / nokv / noogv	nunyi / ninyia (you two)	nunu / nulu (you all)
3rd	bww / mw / nw (he, she, self); object bww vm	bunyi (they two)	bulu / bunu (they)

- **Question words:** hvye / hv yv (who), hugu (what, where), hvkam (how), hwdw (when), akw / hwdw gu (how many).
- **Indefinites:** aku (one, someone; tagged ADJ), kvbi (other), mwnww / mw lww / mvnww (everyone), gwdww (anyone), mvygu (many).
- **Reflexive and emphatic:** atu boongv (oneself), –su (self).

nam has two uses.

- **Immediately after a verb** → PART, the verbal-noun ending (aaku nam “the visiting”).
- **“(to) you”** → PRON, the second-person object pronoun. Dictionary: *Nam ngo me yru kaapa la...* “Yesterday I saw **you**...”. In text: nam ngo kaacham do . “I like you”; ngo nam tuuria dunv kv . “I’m waiting for you.”

nam written for naam “house” is a misspelling and is corrected: nam so aato → naam so aato “come into the house”.

3.4 Tense, aspect and mood

All tense, aspect and mood markers are tagged **PART** and written after the verb as separate tokens. The examples are quoted verbatim from the dictionary.

Meaning	Standard marker(s)	Example (dictionary spelling)	Gloss
Simple present / habitual	do	<i>Ngonu acin dedo.</i>	“We eat food.”
Present continuous	du, dun (1st person), rin	<i>Ngo kitab puri dun.</i>	“I am reading a book.”
Beginning of an action	rap ri	<i>Anga kaprap ri.</i>	“The baby starts crying.”
Present perfect	pv, pvku, ro, roku, ku, puku	<i>Ngo acin dero ku.</i>	“I have eaten.”
Perfect continuous	bv du, bv do	<i>Ngonu seeru loke gyobv do.</i>	“We have been playing since morning.”
Simple past	pa, pato, pvn	<i>Ngo meyru acin depa.</i>	“Yesterday I ate rice.”
Past perfect / sequence	la, tvla, tola	<i>Nam ngo meyru kaapa la agam gu minpa.</i>	“Yesterday I saw you and said something.”
Past continuous	bv / dung + pa / to / do	<i>Ngo puri ridung la doonam e light eero pa.</i>	“The light went out while I was reading.”
Future	re	<i>Aaru nyi ngo Dapo aarey.</i>	“I will come to Dapo tomorrow.”
Future perfect	ro riku, pv riku	<i>Aaru nyi ngo minro riku.</i>	“I will have spoken tomorrow.”
Determination (“surely”)	-ru- + re	<i>Tapa yaaru rey.</i>	“Tapa will surely win.”
Negation	ma, madu, mado	<i>Apu e kaapu maado.</i>	“The flower is not beautiful.”
Command	to	<i>Eero to.</i>	“Go away.”
Request / entreaty	ka, kv, bvkv	<i>Ngam reyla ka.</i>	“Save me.”
Negative command	yo, mabv, ram	<i>Ngam jiyo!</i>	“Don’t give me (food)!”
Inability	kuram, ram	<i>No so menyu ram.</i>	“You can’t do this.”
Question	hai, ri, nvi, la	<i>No so ado hai?</i>	“Have you come here?”
Exhortative (“let’s”)	ju, laju	<i>Ngonyi doobam laaju.</i>	“Let’s sit together.”
Wish	laka	<i>Ngunu am regene si sinso laka.</i>	“May our leader live long.”

Condition. “If” is expressed by bo’lo after the verb: *Nyido oobolo, ngonu eema rey.* “If it rains, we will not go.” By convention bo’lo is tagged **DET** in every use (Appendix G, R12).

Purpose. “So that” is expressed with dubv after the verb: *No ge Oryok e Nyema dube, ngo peeci jido.* “I hide your dao so that it doesn’t get lost.” dubv is tagged **VERB** (§10.2).

3.5 Other word classes

- **Verbal nouns (gerunds):** add -nam to the verb: dvnām eating, kaanam seeing, joknam running. Dictionary: *Jinām e naanam a ale yado* “Giving is better than receiving.” Written as one word these are VERB, or NOUN when the translation is a noun (§10.2).
- **Adjectives from nouns or verbs:** add bv: nyilo bv “manly”, pvcha bv “foolish”, dogum bv “stormy”.
- **Comparison:** do (positive), yaado (comparative), yaachok do (superlative): dictionary *Yaro Oppo si Yapa ge ngam tvsar Yaado* “Yaro’s wine is sweeter than Yapa’s.”
- **Indefinite determiners:** gu, go (“a, one”), tagged DET. aku “one, some” is tagged ADJ.
- **The pointing word nga.** Straight after a noun, nga points back to that noun (“that one”, “its”). It marks the thing being talked about and is tagged **DET**:
 - ginchi naptw nga gamro . “the ginchi’s mouth has been sealed” (nga points to ginchi)
 - morko nga mva ro . “the money is already spent”
 - anga nga so bagha tho . “bring the baby here”

nga is **not** an article. When nga means “me / my” it is a pronoun (PRON): nga issi miang go jito . “give me a few drops of water”; ngo nga gv anyik ... “my eyes ...”.

- **Conjunctions:**
 - “and”: la (between nouns), okv (between clauses)
 - “but”: jeka / rijeka
 - “otherwise”: ho malo / hvbv malo
 - “either ... or”: ya ... ma ... ya

“If” is bolo after the verb (§3.4), or kam linking two clauses.

- **Passive:** the verb is followed by boongv / bu v / angv / nv, and the object takes vm / nga / ngam: dictionary *Kobu angam meki bongē Roji* “The rat was killed by Roji.”
- **Exclamations:** aiya (pain, fear), anyuu / anyung (wow), umbo (agreement), umbonyi / umbonika (thank you), albe (be well), humm (curse), alle (oh no), heloo (hello).

3.6 Numbers and counting

- **Cardinals:** aro (0), akin (1), anyi (2), aom (3), api (4), angu (5), akye / akhe (6), kanw (7), pinw (8), kya (9), ari / aring (10).
 - Tens: acham (a ten), chamkin (10), chamnyi (20), chamom (30)...
 - Hundreds: lvkin (100); thousands: jarkin (1,000); jari (10,000); jari jari (100,000).
- **Joining counts:** gula joins them: arigula akin = 11, chamnyi gula akye = 26.
- **Shape classifiers** replace the initial a of the numeral, so the number word itself tells you what kind of thing is being counted (full table in Appendix B). Examples: pv- round (tape pvkin one pumpkin), tak- flat, bu- long hollow, po- long solid, so- rope-like, bar- coins and discs, bor- paper and cloth, car- small round, do- / dor- animals, lo- days, le- times.
- **Ordinals:** cardinal + jinv (anyi jinv second). For “first”, the word depends on the verb (atu / acho “at first”, kaatu “seen first”).

- **Distributive:** +sak (apple pusak “one apple each”).

Rule

Numerals, classifier-numerals and ordinals are all **NUM**. A classifier-numeral is **one token** (dorkin, pvnyi, lonyi).

PART II

Writing Tagin

CHAPTER 4

The standard orthography (MRST)

The standard spelling is the **Modified Roman Script for Tagin (MRST)**, also known as the GinLish spelling. Every Tagin text prepared under this book **MUST** be written in MRST, or converted to it (Chapter 5), before it is segmented and annotated.

4.1 Letters

a b c h d e g h i j k l m n n g n y o p r s t u v w y

Loanwords may also use f, z, and c without a following h. These letters **MUST NOT** be introduced into native words.

Sound or feature	Standard	Not
the seven vowels	a e i o u v w	
mid front vowel [e]	e	
schwa [ə]	v	ə, and not e (which is [e])
high central vowel [ɨ]	w	v, ɨ, ɨi
velar nasal [ŋ]	ng	ŋ, q
palatal nasal [ɲ]	ny	ɲ, x
palatal affricate [tɕ]	ch	c
long vowel	letter doubled: aa, vv, ww...	a macron (ā) or a colon (a:)
tone	not written	accents, numbers, backticks
letter case	all lowercase , including names and acronyms	capitals
quotation marks	straight " and '	“ ” ‘ ’

Why lowercase?

Tagin has no tradition of capital letters, and different sources capitalise names inconsistently. Writing everything in lowercase removes a large source of spurious variation. Whether a word is a name is recorded by its tags (PROP, and an entity label), not by its first letter.

4.2 Characters that must not appear

4. The standard orthography (MRST)

Found	Replace with
IPA ə / ɛ / ɨ / ɳ / ɲ / ɑ / ɔ	v / w / ng / ny / a / o
Cyrillic or Greek look-alike letters hidden inside Latin words (Cyrillic а, о, е; Greek ο, κ)	the Latin letter
curly quotes	straight quotes
zero-width characters (U+200B, U+200C, U+200D, U+FEFF), musical notes, emoji and other symbols	remove
non-breaking and other special spaces	an ordinary space

Text **SHOULD** be stored in Unicode Normalization Form C (NFC).

4.3 Hyphen

A hyphen joins the parts of **one lexical unit**, and the result is **one token**:

- **Paired and reduplicated compounds:** si-donyi, abu-tanyi, atu-topo, svching-panv, nyiku-mvku, agar-ayar, haalung-haalung; from the Bible review (R43): ajin-anying, rinam-daknam, ridung-singdung, mvdur-ridur, dvnam-tvngnam, pwklin-yarlin, loonyi-loum, ribam-dakbam, rinying-riru, jwngka-daka, mwnglw-kaalwk
- **Fixed names and terms:** donyi-polo, poolu-alu, atar-anyanya
- **Compounds with polo that do not mean “moon”:** tachi-polo (kneecap), yaagam-polo (skeletal), akio-polo, takio-polo (emaciated), rokpu-podo polo (cockerel)

One hyphen in a Tagin word; foreign words keep theirs

Native Tagin words. A hyphen joins exactly **two** parts, so a Tagin word **MUST NOT** contain more than one hyphen. When a native compound has three or more parts, join the first two with a hyphen and write the rest as separate words: rokpu-podo polo (cockerel), not rokpu-podo-polo.

Foreign words MAY contain more than one hyphen. A hyphenated foreign name or term keeps the hyphens of its source, one between each pair of parts, and stays **one token**: majar-i-sarip (Mazar-i-Sharif), pruf-op-uvrk (proof-of-work).

Abbreviations keep their original letters. An abbreviation from English or another language is written as it is in the source language, in lowercase (§4.1), with no hyphens: nft (NFT), upi (UPI), dna (DNA), cd (CD). It **MUST NOT** be respelled with letter names (en-ep-ti, yu-pi-ai, si-di, sidi are wrong).

A double hyphen is a typo. Two hyphens next to each other (--) are never correct inside a word: write one (boosik--bomik → boosik-bomik). No word begins or ends with a hyphen.

Ordinary phrases **MUST NOT** be hyphenated, and an established two-part hyphenated compound **MUST NOT** be split.

CHAPTER 5

Converting other spelling systems to the standard

Tagin has been written in at least four Roman-based systems. This chapter gives the rules for converting each one into MRST. Every example pair below is attested: the source form occurs in the source spelling and the standard form occurs in standard-spelled text.

5.1 From the Tagin dictionary and school-book spelling

Dictionary	Standard	Examples (dictionary → standard)
e [e]	e	<i>seene</i> → <i>seenv</i> (the first <i>e</i>), <i>re</i> → <i>re</i> (future marker)
e [ə]	v	<i>loke</i> → <i>lok v</i> , <i>ge</i> → <i>g v</i> , <i>ke</i> → <i>k v</i> , <i>be</i> → <i>b v</i> , <i>e</i> → <i>v</i> , <i>okke</i> → <i>ok v</i> , <i>ale</i> → <i>a l v</i> , <i>heye</i> → <i>h v y e</i> , <i>ane</i> → <i>a n v</i> , <i>nyime</i> → <i>ny i m v</i>
v [i]	w	<i>elv</i> → <i>v l w</i> (stone), <i>lebv</i> → <i>l v b w</i> (knee), <i>hedv</i> → <i>h w d w</i> (when)
c	ch	<i>acin</i> → <i>a c h i n</i> (food)
am (object marker)	vm (also am)	<i>am</i> → <i>vm</i>

Exceptions. The future marker stays *re*, and the dictionary’s *pe* is standard *pv*.

The mapping is not one-to-one

The dictionary’s *e* is sometimes a true *e*: *seene* “tree” is standard *seenv*, not *svnv* (a different word). Conversion **MUST** therefore consider **every** candidate spelling (each *e* → *e* or *v*, each *v* → *v* or *w*, *c* → *ch*) and keep only candidates that exist in the lexicon:

- exactly one attested candidate → convert (*loke* → *lok v*, *telo* → *t v l o*, *teedo* → *t v d o*);
- several attested candidates → a person decides from context;
- no attested candidate → keep the original and ask a speaker.

An automatic converter **SHOULD** implement exactly this procedure and report the second and third cases for human review.

5.2 From the “Mooya” Bible spelling to the “Mookwng” (standard) spelling

The two Tagin Bible translations follow different spelling traditions. A comparison of their vocabularies shows these systematic differences:

Pattern	How common	Examples (Mooya → standard)
Restore initial h	very common	<i>vbv</i> → <i>hvbv</i> , <i>vjak</i> → <i>hvjak</i> , <i>okv</i> → <i>hokv*</i> , <i>oogv</i> → <i>hoogv</i> , <i>vk v</i> <i>lvgabv</i> → <i>hvk v</i> <i>lvga</i> <i>bv</i>
Split joined phrases	common	<i>ogulvgabv</i> → <i>hugu</i> <i>lvga</i> <i>bv</i> , <i>holvgavbolo</i> → <i>ho</i> <i>lvga</i> <i>bv</i> <i>hvbolo</i> , <i>oguaingbv</i> → <i>hugu</i> <i>aing</i> <i>bv</i> , <i>oguaingnv</i> → <i>hugu</i> <i>aing</i> <i>nv</i>

5. Converting other spelling systems to the standard

Pattern	How common	Examples (Mooya → standard)
v → w	occasional	gvrja → gwrja, lvkcho → lwkcho, lvklin → lwklin, mvnglwkw → mwnglwkw, mvngpa → mwngpa, mvngnam → mwngnam
o → u	occasional	lvko → lvku, nonyi → nunyi, nonu → nunu, oyi → uyi
Vowel and consonant length	occasional	iki → ikki, rigvdogv → rigvdoogv, koching → kooching (also inside longer words: kochinglo → koochinglo), kokwng → kookwng, logo → loogo, lonyi → loonyi, sogv → soogv, dwlo → dww lo, dow → doww v
um → vm	rare	um → vm (the object marker and “that”)

* okv “and then” (CCONJ) is also a real standard word. Restore the h only where the Mooya text means hokv “that one there” (DET), and decide from the parallel Mookwng verse.

The Mooya–Mookwng word list. A list compiled from the two Bibles gives the Mookwng spelling of about 570 Mooya words and is used for conversion. It distinguishes a capitalised, sentence-initial form from the lowercase one (*Vdwlo* “when” → hwdwlo). Where the list contradicts a ruling of this book, the book wins: okv stays okv (R15), the plural marker stays vdw (R16) and um becomes vm (R17); see R40.

5.3 From IPA or linguistic transcription

ə → v, ɪ → w, ŋ → ng, ɲ → ny, tɕ / c → ch, dz → j; a long-vowel mark (:) → doubled vowel; tone marks → removed (tone is not written in MRST).

5.4 Loanwords

Loanwords are spelled the way Tagin speakers pronounce them (more samples in §D.4):

English sound	Tagin	Example
f, ph	p	<i>selfie</i> → selpii, <i>wi-fi</i> → uai-pai, <i>phone</i> → pon
v	b	<i>avatar</i> → abatar, <i>vlog</i> → blog, <i>governor</i> → gobornor
z	j	<i>zoom</i> → juum, <i>zero</i> → jiro
schwa	v	<i>influencer</i> → influensvr, <i>metaverse</i> → metabvrs
th	t or d	<i>hypothesis</i> → haipotvsis, <i>algorithm</i> → elgoridvm
w at the start of a word	u	<i>wi-fi</i> → uai-pai, <i>smartwatch</i> → smartuach
long vowels	doubled	<i>meme</i> → miim, <i>reels</i> → riils
units of measure	as pronounced	<i>kilometre</i> → kilomitar, <i>metre</i> → mitar, <i>feet</i> → pit, <i>miles</i> → mails (R44)
abbreviations	the original letters, lowercase	<i>NFT</i> → nft, <i>UPI</i> → upi, <i>DNA</i> → dna, <i>CD</i> → cd; never letter names such as en-ep-ti (§4.3)
hyphenated foreign names and terms	keep every hyphen	<i>Mazar-i-Sharif</i> → majar-i-sarip, <i>proof-of-work</i> → pruf-op-uvrk (§4.3)

Personal and place names follow the same rules: *David* → devid, *Sophia* → sopia, *Itanagar* → itanagar. Everything is lowercase.

5.5 From Galo spelling (for comparison only)

Galo q = Tagin ng, Galo x = Tagin ny, and Galo double consonants (okkv) correspond to Tagin single consonants (okv). The Galo backtick (a tone mark) has no Tagin equivalent.

Rule

Galo text is **never** converted into Tagin. These correspondences are for aligning the two languages in parallel and comparative work (Appendix J).

CHAPTER 6

Spelling normalization

Normalization removes *inconsistent spellings of the same word*. It never removes genuine variation between speakers (§6.2) and never merges different words that happen to look alike (§6.3).

6.1 Normalization table

Apply these as whole-token replacements: the form must not touch another letter or a hyphen on either side. The table applies to annotated text **and** to any parallel text in the same collection, at the same time.

Variant(s)	Standard	Meaning / condition
um (meaning “that” or “yes”)	vm	Only inside a phrase. The standalone dictionary word um (“hold in the mouth”, “forget”) stays.
vh	v	the subject / topic particle; hv is a different particle and stays
vla	hvla	“saying, speaking”
helo	heloo	“hello”
aalle	alle	“oh no!”
polo (moon, month)	poolo	poolu and poolo are both valid dialect forms; for polo in other compounds see §4.3
poolu alu	poolu–alu	the name of a day
tani	tanyi	the ancestor Tani / Abotani
bongv	boongv	“one who ...”; also in atu boongv “oneself” (R36)
nonu	nunu	“you all”, also at the start of longer words: nonua → nunua, nonuno → nununo (R33)
nonyi	nunyi	“you two” (R34)
lvgabv	lvga bv	“because, for”, also in hugu lvga bv, hvkv lvga bv (R41)
bunugv / hugulo / gvrila / ngogv	bunu gv / hugu lo / gvri la / ngo gv	“their” / “where” / (R41) / “my”
ataranya	atar–anya	“last, end”
ataranyabv	atar–anya bv	“the last one”
tavk	tawk	windmill palm leaf (<i>Trachycarpus fortunei</i>)
adw = plural marker	vdw	e.g. ho vdw hv “those...”, nyii vdw “the people”
adw gv	vdw gv	“their”
adw = “short”	adww	dialect form adwng

Variant(s)	Standard	Meaning / condition
vdwgv / vdwgvlo / vdwlo	vdw gv / vdw gvlo / vdw lo	the plural marker is always its own token
nyi meaning “person, people”	nyii	the tests are in §11.1
nyi adw	nyii vdw	“the people”
nyi mvnwng / nyi mvnww	nyii mvnwng / nyii mvnww	“everyone”
nam meaning “house”	naam	nam so aato → naam so aato “come into the house”
kunyibokar	kunyi-bokar	a paired compound (§4.3)
rokpu-podo-polo	rokpu-podo polo	“cockerel”; one hyphen at most (§4.3)
a native Tagin word with two or more hyphens	first two parts hyphenated, the rest separate words	e.g. rokpu-podo-polo → rokpu-podo polo . Hyphenated foreign names and terms keep all their hyphens: majar-i-sarip, pruf-op-uvrk (§4.3)
an abbreviation spelled with letter names	the original abbreviation, lowercase	en-ep-ti → nft , yu-pi-ai → upi , si-di / sidi → cd (§4.3)
a double hyphen -- inside a word	one hyphen	a typo: boosik--bomik → boosik-bomik (§4.3)
any uppercase letter	lowercase	§4.1

6.2 Dialect variants: keep them as they are

These pairs **MUST NOT** be merged. Give them the **same tags**.

Variants	Meaning	Tags
ngunu / ngulu / ngun / ngul	we	PRON
bulu / bunu	they	PRON
mvnww / mwllww / mwllw / mvnww	everyone	PRON
ngam / ngom	me, mine	PRON
ajin / ajen	friend	NOUN, O
amin / amen	name	NOUN
mintam / mintom / mentom	tell	VERB
poolu / poolo	moon, month	NOUN
nyibu / nyubu	priest, shaman	NOUN, CULT
adw / adwng	short	ADJ
-ng ending vs doubled vowel (kamdung / kamduu, vvming / vvmii, chvbwng / chvbww)	the same word	same tags

6. Spelling normalization

Variants	Meaning	Tags
month names: lwmi / limi, lvchar / lwchar, hilu / hiil, svngtv / svvt, para / pwra, ralw / ralv	Appendix C	NOUN, DATE
season names: dvra / dwra, dvrv / dvrw	winter, summer	NOUN, DATE
the l / n, m / b, d / j, h / y, e / i, o / u and a / o alternations (§2.3)	the same word	same tags

Only merge a pair if the normalization table (§6.1) lists it.

6.3 Pairs that look alike but are different words: never merge

Pair	Difference
nam / naam	verbal-noun ending or “you” (object) vs “house”
nam / nom	nam is the verbal-noun ending or “you”; nom is only “you” (object form)
nyi / nyii	object marker for people vs “person”
jinam / jiinam	“give” vs “plant crops”
go / gu	two different determiners
alo / alu	“bone; there” vs “day”
doodo / doopa	two different tense forms
hvjak / sijak	two different words for “now” (synonyms)
so / soka	“here” vs “at this exact spot”
aachi / achi	two different words
okv / hokv	“and then” vs “that one there”
polo in compounds / poolo	see §4.3
um (hold in the mouth, forget) / vm	see §6.1

PART III

Segmentation

CHAPTER 7

Tokenization

7.1 Rules

Rule	Example
Separate words at spaces.	ngo puri do . → 4 tokens
Every punctuation mark is its own token: , . ? ! ; : () " ' and dashes.	vm , ngo chindo .
An ellipsis ... is three . tokens; the single character ... is one token.	. . .
A hyphenated compound is one token. A native Tagin word contains at most one hyphen; a hyphenated foreign name or term is one token and keeps all its hyphens; an abbreviation is one token in its original letters (nft) (§4.3).	si-donyi
A classifier-numeral is one token (§3.6).	dorkin, pvnyi
Numbers written in digits are one token, including internal separators.	2024, 15, 1,000, 3.5
Scripture references: a single reference is one NUM token. A range is split into NUM + dash + NUM.	20:31; 3:21–4:13 → 3:21 – 4:13; 3:1–12 → 3:1 – 12
Currency amounts and abbreviations stay as written and are one token.	\$5, st., d.c.
Web addresses, e-mail addresses, hashtags and user handles are one token each.	#sidonyi, @tagin

7.2 Particles are separate tokens

Tagin particles follow a word but are **written and tokenized separately**. This is the standard convention.

Written joined (non-standard)	Tokenized (standard)
vdwgv	vdw + gv
vdwgvlo	vdw + gvlo
vdwlo	vdw + lo
atar-anyabv	atar-anya + bv
jisunyi, pwknvyarnvnyi, paulnyi	jisu + nyi, pwknvyarnv + nyi, paul + nyi (R35)
vngjire, minjire	vngji + re, minji + re (future re, R37)
vngroto, linroto	vngro + to, linro + to (command to; ro stays with the verb, R37)
lvgabv, bunugv, hugulo, gvrila, ngogv	lvga + bv, bunu + gv, hugu + lo, gvri + la, ngo + gv (R41)
nonugv, mvnwngnga, naamlo	nunu + gv, mvnwng + nga, naam + lo

Names and nyi. A name and the marker nyi are always two tokens, for people and for God alike: jisu

nyi, pwknvyarnv nyi, kristo nyi, meri nyi (R35).

Lexicalised exceptions. Some words have a particle permanently fused to them and are kept as single tokens: the possessive pronouns ngokv, nokv, noogv; the word jire, which stays one word when it stands alone; and adu roto, which is written as it is (R37). A form is lexicalised when speakers treat it as one word and it appears as a headword or fixed form in the dictionary. When in doubt, split, and record the case for adjudication.

Words kept whole (R42). These look like a word and a particle but are written as one word: hvla, hvkv, hvka, vdw, sinv, nvdw, kanw, gvngv, gvnga, gvnam, alvbv, atugv, arwnglo, nyibv, lakv, sikunv, minsunam, mvngpunam, vvsinam.

Different words joined and apart (R42). These are written joined or apart according to meaning, and both spellings stay as they are: hvbolo / hv bolo, hvnam / hv nam, hogv / ho gv, nyia / nyi a, nyika / nyi ka, nyilo / nyi lo, nyingv / nyi ngv, naama / naam a, soka / so ka, sika / si ka, sima / si ma, sire / si re, kano / ka no, kanam / ka nam, akoka / ako ka, aama / aam a, aadu / a adu, anw / a nw, anyia / a nyia, asia / asi a, aminv / amin v, lookv / lo okv, hala / ha la, yanam / ya nam, gvso / gv so, gvlaak / gv laak, lodvbv / lo dvbv, nyipamv / nyipam v, paadunv / paa dunv, ngonyia / ngo nyia, nonyia / no nyia.

Schemes that attach markers. Some Tagin resources keep case markers *attached* to the word (daporijo-lu). To convert such data, split at the known marker boundary; the entity label stays on the content word and the marker is PART / O (§16).

7.3 Data formats

An annotated token carries three layers: its **form**, its **part of speech** and its **entity label**. Any of the formats below **MAY** be used; they carry the same information and can be converted into one another without loss. Files **MUST** be UTF-8, in NFC, with one sentence per record.

7.3.1 CoNLL-U (recommended for archiving and exchange)

CoNLL-U is the format of the Universal Dependencies project. Each token is one line of ten tab-separated columns; sentences are separated by a blank line and preceded by comment lines. The POS tag goes in column 4 (UPOS), the entity label in column 10 (MISC) as NER=.... Columns that are not annotated hold an underscore.

```
# sent_id = example-001
# text = takar puri do .
# text_en = Takar studies.
1    takar    _    PROP_N    _    _    _    _    _    NER=B-PER
2    puri     _    VERB     _    _    _    _    _    NER=0
3    do      _    PART     _    _    _    _    _    NER=0
4    .       _    PUNCT    _    _    _    _    _    NER=0
```

Recommended sentence comments: sent_id (unique and stable), text (the sentence in MRST), text_en (English translation), and, where known, source, register and variety.

7.3.2 JSON (for software pipelines)

A corpus is a list of sentences; each sentence is a list of tokens; each token has exactly the three fields word, pos and ner.

```
[{"word": "takar", "pos": "PROP_N", "ner": "B-PER"},
 {"word": "puri", "pos": "VERB", "ner": "0"},
```

7. Tokenization

```
{ "word": "do",      "pos": "PART",  "ner": "0"},  
{ "word": ".",      "pos": "PUNCT", "ner": "0"} ]]
```

Write non-ASCII characters directly (not as `\u` escapes). Sentence metadata, when needed, is kept in a parallel list with the same order and the same sentence identifiers.

7.3.3 Parallel text

A parallel (bitext) file pairs each Tagin sentence with its translation, one pair per record, with the columns `tagin` and `english` (and further languages as needed). Spelling rules apply to the Tagin column only. The Tagin column **SHOULD** be identical to the tokenized sentence joined with single spaces, so that annotation and translation can always be re-aligned.

Sentence segmentation

1. **One unit per sentence.** A sentence is a complete utterance. A dictionary headword, a title or a short phrase **MAY** stand alone as a sentence.
2. **Every sentence ends with punctuation.** Use . unless the text is clearly a question (?) or an exclamation (!). Where the source has no final punctuation, add ..
3. **Chants and verses** have no internal full stops, so split them into verse lines:
 - Split at a comma that ends a line. A piece of fewer than three words stays attached to the next line.
 - If there are no commas, split after the line-final verb or particle (laju, laaju, to, riato, bvju).
 - End each line with ..
 - A chant's closing **reciter's name** is its own sentence (takio soki .).
4. **Quoted speech** stays inside the sentence that introduces it, with its quotation marks as separate tokens. A long quotation that contains several sentences is split at its own sentence boundaries.
5. **No exact duplicates.** A released collection **MUST NOT** contain the same sentence twice with the same words *and* the same tags. Genuine repetitions in a text (refrains, formulae) are kept once per collection and their frequency recorded in the metadata if it matters.
6. **No invented entities.** Sentences generated from templates or by machine **MUST NOT** contain invented names of people, places or organisations. Synthetic sentences with real entities **MAY** be included only if they are labelled as synthetic in the data statement (Chapter 20).

PART IV

Part-of-Speech Tagging

CHAPTER 9

The tagset

The part-of-speech tagset is the **Universal POS tagset (UPOS)** of the Universal Dependencies project. Tagin uses **13** of its 17 tags.

Tag	What it covers in Tagin	Examples
PART	case markers, tense / aspect / mood markers, negation, topic, plural, question and discourse particles; the verbal-noun ending <i>nam</i>	<i>lo, gv, nyi, vm, do, re, ma, vdw, hv</i>
NOUN	common nouns, including nouns formed from verbs; names of deities, spirits, festivals and rituals (§10.3)	<i>nyii, naam, isi, donyi, si-donyi</i>
VERB	verbs, including verbs that follow another verb	<i>kaagia, puri, jila, dubv, hvla</i>
PROPN	names of people, places and organisations; names of tribes, communities, languages and religions	<i>takar, dapo, tagin, inglish</i>
PRON	personal, question and quantifying pronouns	<i>ngo, nam, bulu, hvye, mwnww</i>
DET	demonstratives, the pointing word <i>nga</i> , “a / one”, “all”	<i>so, ho, nga, go, mvnwnng</i>
ADJ	adjectives	<i>alv, ai (heavy), nutun</i>
ADV	adverbs	<i>hookv, hvkam, sije</i>
CCONJ	conjunctions	<i>okv, hvbv, jeka, la</i>
NUM	numerals, classifier-numerals, ordinals, digits, scripture references	<i>akin, dorkin, 2024, 3:16</i>
INTJ	interjections, and <i>vm</i> “yes”	<i>alle, heloo, umbonika</i>
PUNCT	punctuation	<i>. , ?</i>
X	foreign or abbreviated tokens that are not Tagin words	<i>of, st., \$5</i>

9.1 Tags that Tagin does not use

UPOS tag	Why it is not used, and what to use instead
ADP	Tagin has no adpositions of its own: the words that translate English prepositions and postpositions are case particles → PART .
AUX	Tense, aspect and mood are carried by particles → PART . A full verb that follows another verb stays VERB .
SCONJ	Clause linkage is marked by particles after the verb (→ PART) or by coordinating words (→ CCONJ).
SYM	Symbols are removed in normalization (§4.2); currency amounts are X ; mathematical signs in running text are PUNCT .

Rule

The tags ADP, AUX, SCONJ and SYM **MUST NOT** appear in Tagin data that conforms to this standard.

9.2 Converting from other tagsets

Older Tagin POS resources used other tagsets. Convert them like this:

Old tag	New tag
ADP (“accusative”, “case”, “preposition”)	PART
CONJ	CCONJ
AUX	PART (a tense marker) or VERB (a verb that follows another verb)
SCONJ	PART (a marker after the verb) or CCONJ (a linking word)
PHRASE	split the phrase into tokens and tag each one
“simple”, “past”, “present”, “tense”	PART for the marker token; VERB for the verb it follows
NNP, NN, VB, JJ, RB and other Penn-style tags	the corresponding UPOS tag, then check the result against Chapter 10

CHAPTER 10

How to decide a POS tag

Ask these questions **in order**, and stop at the first “yes”:

1. **Is it punctuation?** → PUNCT
2. **Is it a foreign or abbreviated token that is not a Tagin word** (of, st., \$5)? → X
3. **Is it a number, a classifier-numeral, an ordinal or a scripture reference?** → NUM
4. **Is it an exclamation, or vm meaning “yes”?** → INTJ
5. **Is it a case, tense, aspect, mood, negation, topic, plural, question or discourse marker** (Appendix A)? → PART
6. **Does it link clauses or nouns** (okv, hvbv, la between nouns...)? → CCONJ
7. **Does it stand in for a person or thing** (I, you, who, everyone...)? → PRON
8. **Does it point to or quantify a noun** (this, that, a, all)? → DET
9. **Is it the name of a deity, spirit, ancestral being, festival or ritual?** → NOUN (and usually CULT, §14.5)
10. **Is it the name of a specific person, place, organisation, tribe, community, language or religion?** → PROP
11. **Is it an action or a state?** → VERB
12. **Does it describe a noun?** → ADJ. **Does it describe a verb, an adjective or a whole clause?** → ADV
13. **Otherwise** → NOUN

Meaning decides

When a translation exists, tag each word by **what it means in the translation**. When there is no translation, use the readings and tests in Chapter 11 and the word lists in Appendix A. If the tag is still uncertain, mark the token as uncertain and send it to adjudication (§18.4). Never guess.

10.1 PART

The full inventory is in Appendix A. Specific rules:

- nyi is the object / recipient marker for **people** (§3.2). It is always PART.
- nam right after a verb → PART (verbal-noun ending). nam meaning “(to) you” → PRON (§3.3).
- vdw → PART, always its own token, NER always O.
- a, ah → PART.
- ai → PART at the end of a sentence; ADJ for “heavy”; NOUN for “elder sister”.
- nvngv → PART when it links clauses; DET when it means “that” right after a noun.
- la → PART, except between two nouns, names or pronouns, where it is CCONJ (§10.6).

10.2 VERB

- Verbs that follow another verb stay VERB: kula, jila, dubv, tvla, hvla.
- Verb + -nam written as one word → VERB. Use NOUN only when the translation is a noun: mvvnam “a lie”, dvnam “food”.
- **Tag by what the translation says was done:** aaku “visited”, vvdum “met”, tvjii “agree”, yaaro “won”, sipa “died”, chvbww “hug”, hingia “like”, aanwk “approach”, aato “please come”, dvki “eat daily”.

10.3 NOUN

- nyii (person), boongv (one who...), donyi (sun), poolu / poolo (moon, month), tawk (palm leaf), ajin / ajen (friend), achi, seeyu (last night), and verbal nouns in -nam when the translation is a noun.
- acham → NOUN when it means “group, subset, cluster”.
- atar-anya → NOUN when it stands alone (“the end”).
- **Names of deities, spirits, ancestral and mythological beings, festivals and rituals are NOUN**, carrying the entity label CULT: si-donyi, abu-tanyi, donyi uyu, doring-chijing, uyu. These names are built from, and often identical to, ordinary Tagin nouns (donyi “sun”, uyu “spirit”), so they share the tag NOUN; the entity label records that they are names.
- **Everyday words are NOUN even when they look like names or are loanwords:** pvga (a hornbill), tador (earthworm), school, sipai (army), hospetal, elekson, sikitwri (secretary), gobornor (governor), yabin (goat, donkey).
- Kinship terms and titles are NOUN (abu, achi, axi), also when they stand before a name (§15.3).

10.4 PROPON

- **Names of people, places and organisations**, including every word of a multi-word name (rajiv gandhi university: three PROPON).
- **Tribe and community names:** tagin, galo, nyishi, adi, apatani.
- **Religion names:** donyi-polo as the name of the faith.
- **Language names** (inglish, french) → PROPON with NER O.
- **Names of foreign peoples and groups** (jius Jews, jentail Gentiles) → PROPON with NER O.
- **Never PROPON:** adjectives like sanga-hara (disappointed), dumda-tabyar (busy), mooya-moi (embarrassed); the interjection umbonika (thank you); verbs like rima, aato, kaaria; paired chant words and ritual compounds that do not name a specific being, place or group: tag these by their meaning, usually NOUN or VERB.

Rule

A PROPON token with entity label O is allowed only for a language name or the name of a foreign people or group. Every other PROPON is inside an entity.

10.5 PRON, DET, ADJ, ADV

- **PRON:** personal pronouns and their object and possessive forms; question words; indefinites (§3.3; §A.3). *nam* “(to) you” → PRON. *nga* “me / my” → PRON; *nga* pointing back to a noun → DET (§3.5). *ngom* is never ADJ.
- **DET:** *ngv*, *so* (this; here), *si*, *ho*, *gu*, *go*, *hokv*, *olo*, *tolo*, *bolo* (also when it means “if” after a verb), *hvkv*, *hoka*, *hogu*, *hvmi*, *hookv*, *boolo*, *golo*, *hvv*, *mvnwng* (all). *go* / *gu* “a, one” → DET.
- **ADJ:** *alv*, *aku* (one, some), *ao*, *kama*, *nama*, *aama*, *nyigam*, *ugu*, *gulo*, *nyuma*, *nyiba*, *bilv*, *arv*; *ai* “heavy”, *nutun* “new”, *dvpu* “tasty”, *agar-ayar* “tired”, *ako* “one (of)”, *angin-ayok* “countless”, *sanga-hara* “disappointed”. *atar-any*a directly before a noun → ADJ (“last”).
- **ADV:** *gubv*, *achvlv*, *alvbw*, *dunv*, *kai*, *soka*, *alvvn*, *mare*, *gunv*, *soka* (at this exact spot), *ribo*, *rimu*, *maadu*, *hookv* “later”, *hvkam* “how”, *takbv* “towards”, *sije* “now”, *yaagu* “a lot”.

10.6 CCONJ, NUM, INTJ

- **CCONJ:** always *okv*, *hvbv*, *hvka*, *jeka*, *nyila*, *namv*, *gvla*, *gola*, *ola*, *jebv*, *gybv*, *nvpvla*; *kam* when it links two clauses (“if”). After a verb, *kam* is VERB.
- **la** between two nouns, names or pronouns → **CCONJ** (*tara la tapa*), but not after a question word. Otherwise *la* → PART.
- **NUM:** numerals, classifier-numerals, ordinals, digits, scripture references; *acham* when it means “ten”.
- **INTJ:** *vm* “yes” (clause-initial and followed by a comma), *heloo*, *alle*, *anyung*, *umbonika*, *haalung-haalung*. INTJ always has NER O.

CHAPTER 11

Words with more than one reading

Tag these words **by meaning**. Use the translation when one exists.

11.1 *nyi* and *nyii*

Reading → spelling and tag	How to tell / examples
object or recipient marker for a person → <i>nyi</i> , PART	It follows the person it marks: <i>nyii mwlww takar nyi kaagia do</i> “everyone likes Takar ”; <i>yaro nyi isi jito</i> “give water to Yaro ”.
person, people → <i>nyii</i> , NOUN	The translation has “person, people, man, everyone, anyone, someone”. Patterns: <i>nyii mwnww / mwlww / mvnwng</i> “everyone”; <i>X nv nyii</i> “a person who is X” (<i>nyikki nv nyii</i> “the wise man”); <i>verb + nv nyii</i> (<i>ngui pyoknv nyii</i> “fish dealer”); <i>ho / so / si / hvkv nyii</i> “that / this / such person”; <i>nyii kvbi</i> “other people”; <i>nyii aku</i> “anyone”; <i>nom nyii kaapv nv ri?</i> “did anybody see you?”

Without a translation

In chants and narratives without a translation, *nyi* that is *not* directly after a person’s name, a pronoun or a noun for a person, and that stands where a noun is expected (as subject, before a determiner, or before *vdw*), means “people”: write *nyii* and tag NOUN.

11.2 Other words

Word	Reading → tag	How to tell / example
vm	object marker, “that” → PART	<i>vm leeto</i> “cuff him”, <i>takar vm</i>
	“yes” → INTJ	clause-initial and followed by a comma: <i>vm , ngo ngo gv napum vm kaagia do .</i> “yes, I like my nose” (the first <i>vm</i> is “yes”, the second is the object marker)
la	“and” between nouns, names or pronouns → CCONJ	<i>tara la tapa</i>
	after a verb or clause → PART	<i>ngo bvmw la doodu</i>
acham	group, subset, cluster → NOUN	<i>nyii acham gu</i> “a group of people”
	ten → NUM	<i>bww ayu acham apo lo naam aapa</i> “he came home at ten”
	sleep → VERB	<i>anga acham toh</i> “make the baby sleep”

Word	Reading → tag	How to tell / example
ai	sentence particle → PART; “heavy” → ADJ; “elder sister” → NOUN	v _{lw} si ai do “the stone is heavy”; ngo gv ai ado “my elder sister has come”
nam	verbal-noun ending → PART; “(to) you” → PRON; misspelled “house” → correct to naam	aaku nam “the visiting” / nam ngo kaacham do “I like you” / naam so aato “come into the house”
nga	“me, my” → PRON; pointing back to the noun before it → DET	nga issi miang go jito “give me some water” / ginchi naptw nga gamro “the ginchi’s mouth has been sealed”
bolo	DET in every use, including “if” after a verb	<i>Nyido oobolo, ngonu eema rey</i> “if it rains, we will not go” (R12)
kam	after a verb → VERB; linking two clauses, “if” → CCONJ	no arw-subu vm jida kam / tara no hvnv nyilo kam ngo aama re
atar-anya	before a noun → ADJ; alone → NOUN	atar-anya hupta “last week” / atuake lokv atar-anya “from first to last”
nvngv	clause linker → PART; “that” after a noun → DET	rokpu nvngv “that rooster”
donyi	the sun → NOUN, O; the Sun deity → NOUN, CULT	donyi v poklin puku “the sun has risen” / donyi uyu “the Sun deity” (§14.5)
poolu / poolo	the moon → NOUN, O; month → NOUN, inside a DATE	no poolu vm kila re? (the moon itself) / limi poolu “the month Limi”
seeyu	“last night” → NOUN, B-DATE	seeyu ngo dakyin la “I was standing the whole night ”. The translation of ngo seeyu iji hamma la as “in a dream” is loose: seeyu is still the time word.
takar	personal name → PROPN; star → NOUN	takar gv amin “the star’s name”
okv / hokv	“and then” → CCONJ / “that one there” → DET	§5.2
ho, nom, ram, siyum, seema	the ordinary word vs a name or homograph	ho chi min siti and nom pen are names (PROP, LOC); siyum “tonight” is NOUN (and also a place name, LOC); seema “dead body” is NOUN
siliguri	a being in chants → NOUN, CULT; the city → PROP, LOC	decide from the passage: chant context → CULT
dugi, dulom and other clan names	clan → PROP, CULT; part of a person’s name → PROP, PER; village → PROP, LOC	takar dugi = B-PER I-PER (§E.2)

PART V

Named-Entity Tagging

CHAPTER 12

The named-entity scheme

12.1 Labels

- **BIO:** B-TYPE begins an entity, I-TYPE continues it, and O marks a token outside any entity.
- An I-X token **MUST** follow a B-X or I-X of the same type.
- Entities are **flat**: they never overlap, never nest, and never cross a sentence boundary. Each token gets exactly one label.
- **Particles are never inside an entity:** `takar nyi` → `takar/B-PER nyi/O`. The only function words that may be inside an entity are those that are part of a name itself, such as the English *and* or *or* in an organisation's official name.

12.2 Types

Type	What it covers	Examples
PER	real people, living or historical, named individually	takar, geeta sharma, jisu kristo
LOC	places and geographic features: villages, towns, districts, states, countries, rivers, lakes, mountains, waterfalls	daporijo, upper subansiri, jerusalem
ORG	institutions, companies, platforms, government bodies, councils, schools, churches as institutions	rajiv gandhi university, facebook
DATE	days, months, seasons, years, times of day, dates, clock times and durations	limi poolu, 2024 anying, aaru
CULT	culturally significant entities: deities and spirits, mythological beings, rituals and festivals, traditional roles, tribes and clans, cultural artifacts; animals and plants in a ritual or symbolic context	si-donyi, nyibu, abu-tanyi, tagin

12.3 Span principles

1. **Mark the whole name, and only the name.** A multi-word name is one span: `upper/B-LOC subansiri/I-LOC`. Particles, case markers and punctuation that follow the name are O.
2. **Longest name wins.** When a name contains another name, mark only the outer one with the outer type: `rajiv gandhi university` is one ORG span, not a PER inside an ORG (§15.1).
3. **Coordinated names are separate spans:** `tara/B-PER la/O tapa/B-PER`.
4. **Adjacent entities of the same type** are separated by starting the second with B-: two people named one after the other without a conjunction are `B-PER B-PER`, not `B-PER I-PER`, unless they are one person's name.
5. **Tag the kind of thing the name names.** A place name stays LOC even when it stands for the

12. The named-entity scheme

people or government of that place; a homograph that names a different thing gets that thing's type (§15.2).

How to decide an entity tag

Ask these questions **in order**, and stop at the first “yes”:

1. **Is the token a particle, pronoun, conjunction, determiner, punctuation or interjection?** → O. (The only exception is a function word that belongs to an official name, §12.1.)
2. **Is it a real person, living or historical?** → PER. Mythological, ancestral and spiritual beings → CULT.
3. **Is it a place?** → LOC. A place that begins an organisation’s name is part of the ORG span when it belongs to the official name.
4. **Is it an institution, company, platform or government body?** → ORG.
5. **Does it locate something in time** (a day, month, season, year, time of day, date, duration)? → DATE.
6. **Does it carry cultural, ritual or symbolic meaning *in this sentence*?** → CULT.
7. **Otherwise** → O.

The context test for CULT

The same word can be CULT in one sentence and O in another. Everyday animals, food, plants, body parts, tools and natural objects are **O**. They become CULT only when the sentence is about a ritual, a sacrifice, an offering, a deity or a sacred practice.

- subu sacrificed at si-donyi → CULT; subu grazing → O.
- donyi uyu “the Sun deity” → CULT; donyi v poklin puku “the sun has risen” → O.
- napum “nose”, ngui “fish” at the market, addin “meat” being cooked → O.

CHAPTER 14

The entity types

14.1 PER

- **Personal names**, with a full name as one span: takar, yania dubi, geeta sharma, tanyi dugi dulom, jisu kristo.
- A reciter's name at the end of a chant (takio soki) → PER.
- A clan name that is part of a person's name → inside the PER span (takar/B-PER dugi/I-PER).
- A title directly before a name belongs to the span when the two together form the name by which the person is known (§15.3).
- **Not PER**: kinship terms on their own (abu father, ai elder sister); groups of people (jius, jentail); the particle after a name; ancestral and mythological beings (→ CULT).

14.2 LOC

- **Places**: villages, towns, circles, districts, states, countries, continents, rivers, lakes, mountains, passes, waterfalls: daporijo, dapo, itanagar, arunachal pradesh, upper subansiri, nyemv kardum, anv nungring, jerusalem, nom pen, nyumchi-nyumtv.
- A multi-word place name is one span, including a generic word that is part of the name (dibang valley, west siang).
- A place that is the site of an organisation but not part of its name is a separate LOC span.
- Gazetteer: §D.3.

14.3 ORG

- **Institutions, schools, colleges, universities, hospitals, banks, government offices, councils, political parties, churches as institutions, companies and online platforms**: rajiv gandhi university, nerist, state bank of india itanagar, tcs, facebook.
- The whole official name is one span, including any place in the name: pasighat/B-ORG municipal/I-ORG council/I-ORG.
- Generic mentions ("the school", "the government") without a name are O.
- Organisation names **MUST NOT** be invented or generated from templates (Chapter 8).

14.4 DATE

Pattern	Tagging
day, time and season words	alu (day), alo, apo (hour, time), aaru / aru, meeru, anying (year), selu (today), aryum (night), svlu → B-DATE

Pattern	Tagging
month name + poolu / poolo	one span: limi/B-DATE poolu/I-DATE (month names: Appendix C)
aari poolu “next month”, mvnying anying “last year”	B-DATE I-DATE
poolu–alu (a named day)	one token, B-DATE
poolu pwom / poolu pwnyi (moon phases used as times)	poolu/B-DATE pwom/I-DATE
year + anying	2024/B-DATE anying/I-DATE
number + tarwk + alu	12/B-DATE tarwk/I-DATE alu/I-DATE
clock time	ayu/B-DATE acham/I-DATE apo/I-DATE (“at ten o’clock”)
number + anying	acham/B-DATE anying/I-DATE (“ten years”)
named days	gumpu alu, chvngtum alu, dootv alu, doogum alu, karbu alu → B-DATE I-DATE
“one day, some day”	only the day word is DATE: alu/B-DATE gunyi/0
“half an hour”	apo/B-DATE tuugu/I-DATE
day-counting numerals	lonyi (two days) → B-DATE
a festival used as a point in time	the festival name stays CULT: si-donyi/B-CULT lo/0 “at Si-Donyi”

Particles between date parts are O, so one date can form several spans: 2024/B-DATE anying/I-DATE gv/0 limi/B-DATE poolu/I-DATE (Appendix F).

14.5 CULT

Include:

- **Tribes, clans and communities:** tagin, galo; the clan names listed in §E.2.
- **Religion, deities, spirits and mythological beings:** si-donyi, donyi-polo, uyu, abu-tanyi, atu-topo, atu-sii, riumching, riumdo, doring-chijing, donyi uyu, and every name in §D.1.
- **Rituals, festivals and practices:** si-donyi (the festival, 3–6 January), mwra-keki, gionam-jarnam.
- **Traditional roles and institutions:** nyibu (priest, shaman), doopam (village council).
- **Ritual-register words** (§D.2): vjwng (ritual “sun”), svnyi (ritual “mithun”).
- **Cultural artifacts:** svching-panv, oppo (as a ritual offering), vjing, bvnyi (lady’s walking stick), oriok.
- **Animals, plants and food in a ritual or symbolic role:** the mithun, pig or fowl of a sacrifice; an offering; an omen animal.

Exclude (O):

- **Standalone donyi, poolu and poolo** in their literal sense (the sun, the moon). They are CULT only in a multi-word divine name (donyi uyu, donyi-polo) or when the sentence addresses or worships them as deities.

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- **Everyday uses of animals, food, plants and body parts** (subu grazing, ngui fish at the market, addin meat being cooked, napum nose).
- **iji** (clothes), **tawk** (palm leaf) and other ordinary objects.
- **Language names** (inglish, french) and **names of foreign peoples** (jius, jentail).
- Kinship terms, interjections and ordinary nouns: alle (interjection), ajen / ajin (friend).

CHAPTER 15

Difficult entity cases

This chapter settles the cases on which careful annotators most often disagree. Each rule is chosen so that two annotators working independently will reach the same answer.

15.1 Names inside names

Only the outermost name is marked, with the type of the whole.

```
rajiv/PROPN/B-ORG  gandhi/PROPN/I-ORG  university/PROPN/I-ORG  
state/PROPN/B-ORG  bank/PROPN/I-ORG  of/PROPN/I-ORG  india/PROPN/I-ORG  
↪ itanagar/PROPN/I-ORG
```

rajiv gandhi is not marked PER and india is not marked LOC inside these names. A festival named after a deity, or a village named after a clan, is marked with the type of the festival (CULT) or the village (LOC).

15.2 Metonymy and homographs

- A **place name keeps LOC** when it stands for the people, the government or a team of that place: itanagar “the government in Itanagar” → LOC.
- An **organisation name keeps ORG** when it stands for its building or campus: “at the hospital *X*” → ORG.
- A **homograph** that names a different thing is tagged by what it names: siliguri as a being in a chant → CULT, the city → LOC; takar “star” → O, the person → PER; seema “dead body” → O, the personal name → PER.

15.3 Titles, kinship terms and forms of address

- A **kinship term or title on its own** is NOUN, O: abu “father”, achi “brother”.
- A title **directly before a name**, where the two together are how the person is known, is part of the PER span. The title keeps its POS (NOUN) and takes the B-PER label; the name continues the span: in Galo, axi/NOUN/B-PER aqam/PROPN/I-PER.
- A title that merely describes the person (“the teacher Takar”) is outside the span: takar alone is PER.
- Honorific names of deities that begin with a kinship word (abu-tanyi, atu-topo) are one CULT token.

15.4 Coordination and lists

Each name in a list is its own span, even when the names share a word:

15. Difficult entity cases

```
tara/PROPN/B-PER la/CCONJ/O tapa/PROPN/B-PER  
upper/PROPN/B-LOC subansiri/PROPN/I-LOC ,/PUNCT/O west/PROPN/B-LOC siang/PROPN/I-LOC
```

When a shared word is not repeated (“Upper and Lower Subansiri”), mark the complete name as one span and the incomplete one as a separate span: upper/B-LOC la/O lower/B-LOC subansiri/I-LOC.

15.5 Particles and case markers after names

Case markers, the plural *vdw* and all other particles are O, even when they are written joined to the name in the source. Split them first (§7.2):

```
daporijo/PROPN/B-LOC lo/PART/O (not "daporijolo")  
takar/PROPN/B-PER nyi/PART/O
```

15.6 Foreign, English and code-mixed names

- Names from English, Hindi or any other language are tagged exactly like Tagin names, after respelling in MRST and lowercasing (§5.4).
- Names in scripture follow the same rules: *jisu kristo* → PER; *jerusalem* → LOC. Scripture references (3:16) are NUM, O.
- Names of peoples and groups (*jius* Jews, *jentail* Gentiles) and language names (*inglish*) are PROPN, O.
- Brand names used as ordinary words (“a Facebook post”) are still ORG when they name the company or platform.

15.7 Names used as ordinary words

A name used to describe rather than to refer is not an entity: “a Takar-like person”, “Tagin-style houses”. The tribe name in such an adjectival use is PROPN, O. A deity or festival mentioned only as a word being explained (“the word *si-donyi* means...”) is still CULT, because it names the being or festival.

15.8 Dates without a fixed point

Relative and indefinite time expressions are DATE when they contain a time word (*aaru* “tomorrow”, *aari poolu* “next month”). Only the time word is marked; quantifiers and particles around it are O: *alu/B-DATE gunyi/O* “one day”. Words for frequency without a time word (“often”, “always”) are ADV, O.

15.9 When the rules are not enough

If a case is not covered, choose the reading that the translation supports, mark the token as uncertain, and send it to adjudication (§18.4). Adjudicators record the ruling in the project’s decision log; recurring rulings are proposed for the next edition of this book (Chapter 22).

CHAPTER 16

Crosswalk to other entity schemes

Tagin data annotated under other entity schemes can be mapped to this one. The fine-grained ethno-cultural scheme of Tagin-EtLiNER (Appendix K) maps as follows:

Fine-grained type	This book
MYTH (mythological and spiritual beings)	CULT
RITP (ritual practices and festivals)	CULT
TROL (traditional roles)	CULT
ARTI (cultural artifacts)	CULT
SOC (clans, hereditary lineages)	CULT
ANM (animals and birds)	CULT in a ritual or symbolic context; O otherwise
GPE (towns, states, countries)	LOC
PER, LOC, ORG, DATE	unchanged

The coarse CoNLL-2003 scheme maps as: PER → PER, LOC → LOC, ORG → ORG, MISC → CULT when the entity is cultural or religious, otherwise O (MISC entities such as nationalities and events have no type here). Scripture, time and number annotations from other schemes map to DATE only when they locate something in time.

Tokenization difference. Schemes that keep case markers attached (daporijo-lu = B-LOC) are converted by splitting off the marker, keeping the label on the content word, and tagging the marker PART / O. To convert back, join the two and keep the content word's label.

Going to a finer scheme. Because CULT is a union of finer types, converting *from* this book to a fine-grained scheme needs a human (or a gazetteer, Appendix D) to choose the fine type. The mapping above is lossless only in the direction shown.

PART VI

Making and Maintaining Data

CHAPTER 17

The annotation workflow

Every new Tagin dataset passes through the same stages. Skipping a stage **MUST** be recorded in the data statement.

Stage	What to do	Output
1. Consent and scope	Agree with speakers and community bodies on what is recorded, how it will be used and shared, and at which access level (Chapter 23).	a consent record
2. Collection	Record or collect text with metadata: speaker (pseudonymised), village or region, register (everyday, ritual, scripture, school, media), date, source.	raw text + metadata
3. Transcription	Write in the standard spelling (Chapter 4). If the source uses another system, convert it (Chapter 5).	standard text
4. Normalization	Apply the normalization table (§6.1) and the character rules (§4.2); review every item a converter could not decide.	normalized text + review list
5. Segmentation	Tokenize (Chapter 7) and split sentences (Chapter 8).	tokens
6. Translation	Add an English translation for every sentence whenever possible. The translation decides many tags (Chapter 10).	parallel text
7. Pre-annotation	Optional: a lexicon lookup or a model proposes tags (§19.3).	draft tags
8. Annotation	At least two trained annotators tag independently, following Parts IV–V.	two annotation sets
9. Adjudication	An adjudicator resolves every disagreement and records the reason (§18.4).	gold tags + decision log
10. Validation	Run every check in Chapter 21; everything must pass.	validation report
11. Community review	Speakers review a sample, and all sacred or sensitive material, before release.	review notes
12. Release	Version number, data statement (Chapter 20), licence, access level, edition of this book.	released dataset

Gold data

Only data that has passed stages 8–10 may be called *gold*. Data annotated by one person, or by a model and checked by one person, is *silver* and **MUST** be labelled as such.

CHAPTER 18

Annotators, training and adjudication

18.1 Who annotates

Annotators **SHOULD** be fluent Tagin speakers who read the standard orthography confidently. At least one person in every annotation team **MUST** be a native speaker. Linguists and engineers who are not speakers **MAY** annotate POS in material with reliable translations, but their work **MUST** be adjudicated by a speaker.

18.2 Training and qualification

1. **Study.** The annotator reads Parts I–V of this book and the quick reference card (Appendix H).
2. **Guided practice.** The annotator tags at least 100 sentences alongside an experienced annotator, discussing every difference.
3. **Qualification test.** The annotator tags 200 sentences of existing gold data, drawn from several registers, without help.
4. **Threshold.** The annotator qualifies with POS accuracy $\geq 95\%$ and entity F1 $\geq 85\%$ against the gold. Otherwise, the errors are reviewed and a new test set is used.
5. **Refresher.** After every new edition of this book, and after any break of more than three months, annotators review the edition history (Appendix L) and tag a short calibration set.

18.3 Independent double annotation

Two annotators tag the same sentences **without seeing each other's work**. Pre-annotations, if used, **MUST** be shown to both or to neither. Each annotator **MAY** mark any token as *uncertain* and add a short note; uncertainty marks are part of the output and go to adjudication.

18.4 Adjudication

An adjudicator — an experienced annotator who is a native speaker, and who did not annotate the sentence — resolves every disagreement and every uncertainty mark.

1. Look up the rule in this book that covers the case, and apply it.
2. If the book does not decide the case, use the translation and ask a second speaker when needed.
3. Record every decision in the project's **decision log**: the sentence identifier, the token, both annotations, the ruling, and the reason (a section of this book, or the evidence used).
4. When the same kind of decision recurs, propose it as a rule for the next edition of this book (Chapter 22).

Rule

Adjudication **MUST NOT** be done by majority vote alone. Every ruling cites a rule of this book or

the evidence on which it rests.

CHAPTER 19

Quality targets and agreement

19.1 Targets

Measure	Target
POS agreement between annotators (Cohen's κ , per token)	≥ 0.85
NER agreement between annotators (span-level F1, exact match)	≥ 0.80 overall; ≥ 0.70 for CULT
Share of tokens still marked uncertain after adjudication	$\leq 1\%$
Validation checks passed (Chapter 21)	100%
Sentences with a translation	$\geq 60\%$

19.2 Measuring agreement

- **When.** Measure agreement on a pilot of at least 500 sentences before annotating a whole new domain or register, and on a random 10% of all later batches.
- **POS.** Compute Cohen's κ over all tokens, and also per tag, to find the tags that need clearer rules. With more than two annotators, use Krippendorff's α .
- **NER.** Compute precision, recall and F1 of one annotator's spans against the other's, counting a span as matched only when its boundaries *and* type are identical. Report the scores per type as well as overall.
- **Report.** Publish the agreement scores, the sample size and the method in the data statement (Chapter 20).

When agreement is below target, list the disputed cases, decide them, extend the project's guidelines (and, where they are general, this book), and re-annotate the affected material. Do not raise agreement by simply retraining annotators on the disputed sentences.

19.3 Model-assisted annotation

Tag suggestions from a lexicon, a trained tagger or a large language model **MAY** be used to speed up annotation, under these conditions:

1. Suggestions are shown as *suggestions*. Every token is still checked by a person.
2. Agreement is measured on a sample annotated *without* suggestions, to detect annotators simply accepting what the model proposed.
3. The data statement records which model or lexicon was used, on how much of the data, and how suggestions were presented.
4. Machine-generated *text* (as opposed to machine-suggested *tags*) is synthetic and **MUST** be labelled as such (Chapter 8).

5. Sacred or restricted material **MUST NOT** be sent to an external service without the permission required for it (Chapter [23](#)).

Data statement

Every released Tagin dataset **MUST** be accompanied by a data statement. Copy the headings below and fill them in.

1. **Title, version and identifier.** A stable name, a version number, a release date and, where possible, a persistent identifier.
2. **Curation rationale.** Why the data was collected and what it is for.
3. **Language variety.** Tagin (tgj); region (upper / lower, villages); register (everyday, ritual, scripture, school, media); the spelling system of the source and how it was converted.
4. **Speakers and authors.** Number, age range, gender distribution, villages. Pseudonymise; never publish personal identifiers without consent.
5. **Annotators.** Number, whether they are native speakers, training received, qualification scores, agreement scores (Chapter 19).
6. **Text characteristics.** Genres, time period, size (sentences and tokens), share of sentences with translations, share and kind of synthetic content.
7. **Annotation process.** Stages followed and any skipped (Chapter 17); pre-annotation tools and models used (§19.3); gold or silver status.
8. **Standard.** “Annotated according to *The Tagin Annotation Rule Book*, edition 5.0”, plus any project-specific deviations, each with its reason.
9. **Provenance, licence and access.** Sources and permissions; licence (for example CC BY-SA 4.0 for community-approved open text); access level (open, community or restricted, §23.2); restrictions on sacred content; contact for take-down requests.
10. **Known limitations.** Varieties or registers that are missing or under-represented, and any systematic uncertainties.

CHAPTER 21

Validation checklist

Data that conforms to this standard passes every check below. The checks are mechanical and **SHOULD** be run automatically on every file before release; any failure is fixed, not waived.

21.1 Format

- ☐ Every token has exactly a form, a POS tag and an entity label; the form is non-empty and contains no spaces.
- ☐ The file is UTF-8 in NFC; sentence identifiers are unique.
- ☐ No native word token contains more than one hyphen; only hyphenated foreign names and terms (majar-i-sarip) may contain several. Abbreviations are in their original letters, never spelled with letter names (nft, not en-ep-ti). No token contains a double hyphen --, and none begins or ends with a hyphen (§4.3). A hyphen standing alone as a PUNCT token, as in a scripture range, is allowed.
- ☐ Every sentence ends with a PUNCT token.
- ☐ No exact duplicate sentences (same forms and same tags).

21.2 Characters and spelling

- ☐ No uppercase letters. No IPA letters, no Cyrillic or Greek look-alikes, no curly quotes, no zero-width or other invisible characters (§4.2).
- ☐ None of the non-standard spellings in the normalization table remain: vh, vla, helo, aalle, tani, bongv, ataranya, tavk, kunyibokar, adw as the plural, vdwgv, vdwlo, vdwgvlo, standalone polo, um inside a phrase, the sequence poolu alu, nonu and words beginning with it, nonyi, and the joined forms lvgabv, bunugv, hugulo, gvri la and ngogv.
- ☐ No name is joined to nyi (jisunyi, pwknvyarnvnyi, paulnyi), and no verb is joined to the future re or the command to (vngjire, vngroto), apart from the exceptions in §7.2.

21.3 Tags

- ☐ The POS tag is one of the 13 tags of Chapter 9; ADP, AUX, SCONJ and SYM never occur.
- ☐ The entity label is 0 or B- / I- + PER, LOC, ORG, DATE or CULT; every I-X follows B-X or I-X.
- ☐ PUNCT, PART, PRON, CCONJ, DET and INTJ tokens have entity label O (except function words inside an official name, §12.1).
- ☐ X contains only foreign or abbreviated leftovers.
- ☐ PROP N with entity label O is only a language name or the name of a foreign people or group (§10.4).

21.4 Word-specific rules

- ☐ `nyi` is always PART; `nyii` is always NOUN; `naam` is always NOUN.
- ☐ `vdw` is always PART with entity label O.
- ☐ `okv`, `hvbv`, `hvka`, `jeka` and `nyila` are always CCONJ.
- ☐ `kula`, `jila`, `dubv` and `hvla` are always VERB.
- ☐ `la` is CCONJ only between two NOUN, PROPEN or PRON tokens; otherwise PART.
- ☐ Standalone `donyi`, `poolu` and `poolo` with entity label CULT are reviewed by a person (§14.5).
- ☐ Everyday nouns with entity label CULT (animals, food, body parts) are reviewed by a person against the context test (Chapter 13).

The last two checks produce a review list rather than an automatic failure: a person confirms each item or corrects it.

CHAPTER 22

Maintaining this standard

A standard is only useful if it is stable, and only trusted if it can be corrected. This chapter describes how the book changes.

22.1 Editions

Editions are numbered *major.minor*.

- A **major** edition changes what a correct annotation is: a tag is added, removed or redefined, or a rule is reversed. Data annotated under an earlier major edition **SHOULD** be updated before it is combined with newer data.
- A **minor** edition clarifies or extends the rules without making earlier correct annotations wrong: new examples, new word lists, new sections for cases the book did not cover.
- Corrections of typing errors that change no rule do not change the edition number.

Every change is listed in the edition history (Appendix L), and every rule decided on evidence is recorded in the decision register (Appendix G).

22.2 Proposing a change

Anyone may propose a change: a speaker, a teacher, an annotator, a scholar. A proposal states:

1. the rule or case concerned, with the section of this book;
2. real examples, with translations, showing the problem;
3. the proposed rule; and
4. the evidence: speakers' judgements, dictionary entries, translations, published work.

22.3 Deciding a change

The maintainer reviews the proposal with native-speaker reviewers. Proposals that touch sacred or culturally sensitive matters are also reviewed by the relevant community authorities. The decision, with its reasons, is added to the decision register. When speakers and the book disagree, the book is corrected.

22.4 Applying a change

An adopted rule is applied to all maintained data *and* to its parallel text at the same time, and the new edition is released with its history entry. Datasets state the edition they follow (Chapter 20), so users can always tell which rules a file obeys.

PART VII

Stewardship and Ethics

CHAPTER 23

Caring for Tagin data

Tagin speech and text belong first to the Tagin people. Anyone creating or using Tagin data **SHOULD** follow the **CARE principles** for Indigenous data governance (Collective benefit, Authority to control, Responsibility, Ethics) together with the **FAIR principles** (Findable, Accessible, Interoperable, Reusable). FAIR makes data usable; CARE makes sure that its use serves the people it comes from.

23.1 Principles

1. **Informed consent.** Record, transcribe and publish only with the speaker’s informed consent, given in a language the speaker understands. For communal knowledge (myths, genealogies, place lore, ritual texts) also seek community consent, for example from the Tagin Cultural Society or village elders.
2. **Sacred and ritual material.** Chants of the nyibu, invocations and restricted ritual knowledge may not be meant for public release. Mark such material as restricted (§23.2) and obtain explicit permission before publishing it or training models on it. Never mock, alter or remix sacred texts.
3. **Personal information.** Pseudonymise speakers. Do not publish phone numbers, addresses, health information or private conversations.
4. **Attribution.** Credit speakers, translators and annotators, with their consent, in every release.
5. **Benefit to the community.** Return resources to the community in usable form: dictionaries, school materials, keyboards, spell-checkers, speech tools, and the models trained on community data.
6. **Respect for variation.** No variety is “wrong”. Upper and lower speech, the everyday and ritual registers, and the forms of older and younger speakers are all Tagin.
7. **Honesty about synthetic data.** Label machine-generated and template sentences clearly; never present them as natural speech.

23.2 Access levels

Every text and recording **MUST** carry one of three access levels in its metadata:

Level	Meaning
open	May be published and used by anyone under the dataset’s licence, including for training publicly released models.
community	May be shared with and used by the Tagin community and by projects that have the community’s permission; not for public release.
restricted	Sacred, ritual or personal material. Kept only with the permission of the knowledge holders, used only for the purposes they agree to, and never included in public datasets or public models.

Where the community wishes, cultural provenance and protocols **MAY** also be expressed with Traditional Knowledge (TK) and Biocultural (BC) Labels from the Local Contexts initiative.

23.3 Take-down and correction

Every release names a contact. A request from a speaker, a family or a community authority to correct or withdraw material is acted on promptly: the material is removed from the next release and from any public copy under the project's control, and the removal is recorded. Models trained on withdrawn restricted material **SHOULD** be retrained or withdrawn.

PART VIII

Appendices

APPENDIX A

Grammatical words

All forms are in the standard spelling. These lists are the closed classes of Tagin: when a token is one of these forms, its tag is found here unless Chapter 11 gives it another reading.

A.1 Case, plural and discourse particles (PART)

Form	Function
lo	location “in, at, on”
gv, gvlo	possessive “of, ’s”
nyi	object / recipient marker for people
bv	“towards, for”; adverb-former (“-ly”); continuous aspect
hv	topic marker
vm, am	object / recipient marker for non-humans; “that”
nv	“one who, that which” (relative)
v	subject / topic marker
lokv, lookv	“with, by means of, from”
kv	possessive; “then, okay”
bvlo, tvlo, vlo	direction + location: down / up / level
vdw, vdwv, vdwa, gwdw, wdw	plural
ka	“also, too, even”; request
lvga, followed by bv: lvga bv	“because, for” (two tokens, R41)
nam	verbal-noun ending (after a verb; “you” is PRON, §3.3)

A.2 Tense, aspect, mood and polarity (PART)

Forms	Meaning
do	present, habitual
du, dun, rin	continuous
ro, roku, ku, puku, pvku, pv	perfect
pa, pato, pvn	past
la, tvla, tola	sequence, past perfect (tvla as a full verb after a verb is VERB)
re	future

Forms	Meaning
riku	future perfect
ma, madu, mado, mabv, maabv	negation
to	command
ka, kv, bvk	request
yo, ram, kuram	negative command, inability
ju, laju	“let’s”
laka	wish
hai, ri, nvi, la	question
ai, tho, ha, ham, yu, ya, ke, na, da, a, ah	sentence-final particles

A.3 Pronouns (PRON)

ngo, no, nam (you, object), bww, mw, nw, ngunu, ngulu, ngun, ngul, nunu, nulu, bulu, bunu, ngunyi, nunyi, ninyia, bunyi, ngom, ngam, nga (me, my), nom, ngvm, ngokv, nokv, noogv, ninyigv, hugu, hvyv, hvye, hwdw, akw, kvbi, mwnww, mwlww, mwlw, mvnww, gwdww, mvygu, alakv, naakum

A.4 Determiners (DET)

ngv, nga (pointing), so, si, ho, gu, go, mvnwng, mwnwng, hokv, olo, tolo, bolo (also “if” after a verb), boolo, golo, hvjak, hvkv, hoka, hogu, hvgu, hvmi, hookv, nyepak, sokv, hvv, nvngv (“that”, after a noun)

A.5 Conjunctions (CCONJ)

okv, hbbv, la (between nouns, names or pronouns), namv, gvla, jeka, rijeka, hvka, ola, jebv, gbbv, gola, nyila, nvpvla, kam (linking two clauses)

A.6 Interjections (INTJ)

vm (yes), heloo, alle, anyung, anyuu, umbonika, umbonyi, umbo, haalung–haalung, aiya, albe, humm

APPENDIX B

Numerals and classifiers

Value	Cardinal	Value	Cardinal	Value	Cardinal
0	aro	6	ake / akhe	20	chamnyi
1	akin	7	kanw	30	chamom
2	anyi	8	pinw	100	lvkin
3	aom	9	kya	1,000	jarkin
4	api	10	ari / aring / chamkin	10,000	jari
5	angu	a ten	acham	100,000	jari jari

Joining numbers: gula (“plus”): arigula akin 11, aring gula api 14, chamnyi gula ake 26, lvgula champi 140.

Classifier prefixes. The classifier replaces the initial a of the numeral; the whole word is **one token, NUM**.

Prefix	What it counts	Example
dor- / do-	animals	subu dorkin one mithun
pv-	big round things (stones, fruit, balls)	tape pvkin one pumpkin
car-	small round things (beads, seeds, insects)	tasi carkin one bead
tak-	flat things (plates, planks, bread)	roti taknyi two roti
bu- / bung-	long hollow things (pipes, bamboo)	heju buring ten bamboos
po-	long solid things (logs, batteries)	pokin, ponyi
so-	rope-like things	sokin, sonyi
bar-	discs, money, swords	morko barnyi two rupees
bor-	paper, leaves, mats, cloth	oku borom three leaves
rum-	kinds of clothes	ej i rumkin one cloth
kar-	coils	kokar karpi four coils
paa-	packets	biscuit paangu five packets
lo-	days	lonyi two days
le-	times	lekin once, lenyi twice

Ordinals: cardinal + jinv (anyi jinv 2nd, chamnyi gula aom jinv 23rd). For “first”, the word depends on the verb: atu / acho (at first), kaatu (seen first), dootu (sat first).

Distributive: +sak (nyii asak each person).

Tagging. All of these are NUM. A numeral inside a date or time expression is part of the DATE span

(§14.4); otherwise its entity label is O.

APPENDIX C

Calendar

C.1 Months and seasons

The standard form comes first; the other forms are regional variants of the same name (§6.2) and are kept as written. All are NOUN. A month name followed by poolu / poolo (“month”) forms one DATE span: lwmi/B-DATE poolu/I-DATE.

Month	Standard	Regional variants
January	rajo	—
February	lwmi	limi
March	lvkang	—
April	lvchar	lwchar
May	hilu	hiil
June	talū	—
July	svngngu	—
August	svngtv	svvt
September	para	pwra
October	lwbw	—
November	ralw	ralv
December	ratv	—
winter	dvra	dwra
spring	modw	—
summer	dvrw	dwrw
autumn	jedw	—

Some villages use further month names of their own, such as tenlv, lwkw and seeu. They are tagged by the same rule: NOUN, and DATE together with a following poolu / poolo.

C.2 Time words

Each of these begins a DATE span (§14.4):

alu / alo (day), apo (hour, time), anying (year), aaru / aru (morning, tomorrow), meeru / meyrū (yesterday), seeru (morning), selu (today), seeyu (last night), aryum (night), ayu (midnight), svlu, loodak (daily), kamchi (early morning), atu-aki, svlu-meeru; the named days gumpu alu, chvngtum alu, dootv alu, doogum alu, karbu alu, poolu-alu.

APPENDIX D

Gazetteers

D.1 Deities, spirits and mythological beings

Every name in this list is tagged **NOUN** with the entity label **CULT** (§10.3, §14.5). The names are given here as they are usually written in English-language sources, with capitals. In annotated text they are written in the standard spelling, in lowercase, with paired names joined by a hyphen (abu-tanyi, si-donyi, donyi-polo).

These descriptions record beliefs held by Tagin people. They are presented with respect, as community knowledge, and are not to be used out of context.

#	Name	Description
1	Abotani, Abotanyi	The first human and divine forefather of the Tagins and of the other Tani tribes of Arunachal Pradesh. Also known as Nibo.
2	Ane-Yape, Aneyapv	The divine foremother: the holy mother of the Tagins.
3	Abo-Apa Uyu	The family deity of the ancestral spirits, who watch over the well-being of their descendants.
4	Bibu-Libu	Deity of well-being.
5	Bootar Wiyu	The lord who protects a person from evil spirits and from harm done by others in society.
6	Bootee Radey	The god of wealth in all its forms, whose blessing helps to relieve poverty.
7	Buru	Water deity.
8	Chene Mayu Wiyu	The goddess who cares for the unborn child in the mother's womb. She may also punish it.
9	Changbar's Taming (nyemv) and Dobar's Yami (nyilo)	A mythological couple who eloped and married. Their child was miscarried, and this is held to be the origin of most impurity on earth.
10	Chi Wiyu	The spirit that can complete a newborn's body if anything is lacking at birth. The priest seeks its blessing through a ritual.
11	Chingtum Bootee	The lord of the home, whose blessing and love bring prosperity to the household and its members.
12	Chipo	The mother of spirits, who arose from Sechi.
13	Chungne	The female form of an angel, from whom the Anyi Chungne Ponung troupe descends.
14	Damma-Kano	Spirits that bring poverty, famine and misfortune.
15	Danne Uyu	The goddess of crops, whose blessing brings bumper harvests.
16	Didik Wiyu	The spirit that can cause and cure stomach ailments.
17	Diri-Ane	Deity of grain.

#	Name	Description
18	Doji	Land deity.
19	Donyi Gompa	The abode where human souls are kept under the supreme spirit. Rituals are performed to remain safe there and to keep the family growing rather than dying out.
20	Donyi Wiyu	The sun, revered as a powerful female deity with great power over all.
21	Donyi-Polo	The Sun and the Moon together as the supreme divinity, source of well-being and prosperity for all. From it comes the Donyi-Polo faith.
22	Doley Uyu	A spirit lord that easily afflicts people with illness, mainly fever.
23	Doring Chijing	A spiritual person who could foretell the future, and from whom divination by chicken liver and eggs originated. The lord of wisdom.
24	Dugu Loyii Tamit	The goddess of rice, once married to Abotanyi.
25	Gangding Wiyu	A holy spirit that protects health by hiding a person's soul in a safe place. In its ritual the person hides from evil spirits by changing dress: men wear women's clothes and women wear men's.
26	Ghiri Uyu	The spirit that drives a person to murder.
27	Gungte Uyu	The spirit that brings great rituals and great success. It came to earth from heaven with the two sisters Hoik and Huji (local vegetables).
28	Gute-Gamro	Gods of prosperity.
29	Hammi Panam Wiyu	The lord who, when offered a ritual sacrifice, curses thieves and wrongdoers.
30	Heme Galing	A holder of spiritual power whose gifted soul can fly like a bird.
31	Hibu-Tamu	God and goddess of creation; spiritual supporters of a person's soul for a healthy and prosperous life.
32	Hilo Wiyu	The lord of many of life's capabilities, to whom the mithun (<i>Bos frontalis</i>) was sacrificed for the well-being of the people concerned. Invoked for the welfare of the family.
33	Hite Dekru	A sharp-pointed, poisonous evil spirit.
34	Huru-Tamu	A dangerous spirit.
35	Inju	A female deity who dwells on earth and stays with humans and animals.
36	Ishing and Ili	Two spirits held responsible for miscarriage and stillbirth.
37	Issi Wiyu	The spirit of water.
38	Jaring Wiyu	The deity of the neighbourhood.
39	Jedi-Ane	The mother of the soil: the deity of everything that grows on land, such as food grains.
40	Jelo, Galing	The god who controls the power of thought and speech. Through this spirit (also called Jalo) the priest foretells the future, chanting while beads turn on a red thread. It also brings success in trapping, marriage and similar matters.
41	Jiit-Ane	The youngest daughter of Donyi (the sun) and wife of Abotani. She is believed to have given people their domestic animals and food grains.

#	Name	Description
42	Kiasing Wiyu	A holy spirit that helps a person's soul to become a priest or a hunter, and to gain wealth and other valuables.
43	Kibu-Libu, Hakki Ramki	The spirit of a close relative after his or her death.
44	Kibu-Ryabu, Kibo-Ramlo	A race of spirits that counters and punishes irresponsible human actions, keeping the balance and flow of life.
45	Komdu	One of the spirits that help in easy childbirth.
46	Kungbu Dekne Ningpho	A small evil spirit associated with bitterness and poisonous substances.
47	Kuryum-Kulu	The creator of Donyi (the sun), Polo (the moon) and Hai (water, rivers).
48	Labok Wiyu	A spirit that helps a family to grow. In its ritual a bamboo tube is struck; a loud sound foretells more family members in the near future.
49	Langge-Langya	A spirit that strikes with a stick those who do not observe taboos.
50	Lengri Niki	A deity honoured by the priest as his protector, believed to dwell in the priest's house.
51	Likte and Rote	Evil spirits dwelling in the sky and watching all human beings on earth. They generally cause bodily pain.
52	Lika	A minor spirit of Doli that causes mild communicable diseases.
53	Loi Wiyu	The spirit, and ritual, of healthy digestion.
54	Longre Yullo	The guardian deity of righteousness, invoked especially during the making of Tachi-Tha and other rites.
55	Mangjing Yapom	The lord or king of the mountain.
56	Mitung Wiyu	Mostly evil spirits, raised by priests, that bring bad luck and decline to a family.
57	Nepo	The spirit whose action causes stomach ailments such as pain, diarrhoea and indigestion.
58	Niya Loma	The lord of the smithy, maker of beautiful and deadly weapons and of other useful things.
59	Nikum Pokom	The lord of spiritual knowledge.
60	Nioro Uyi	The lord or king of the forest.
61	Niging Nilo, Channe Panne	The god of orators, through whom one becomes a good speaker.
62	Nikiok-Nyibu Uyu	The lord who gives some people the gifts of a mediator: to chant and to become a priest.
63	Nyida Uyu	The lord who enables a person to arrange great marriage alliances.
64	Niki-Jilo	The spirit that guides the priest (nyibu) safely and gives him strength.
65	Niding-Babu Botar Wiyu	The lord whose blessing brings wealth and success in hunting. Observed by men and women, with various taboos, also for fertility.
66	Nitekoku	A spirit believed to help in conception and birth.
67	Nite-Nigam Uyu	Holy spirits that help a person to become wealthy.

#	Name	Description
68	Nyia (Sotung Wiyu)	A lord who, when offered a sacrifice in a case of elopement, kills the guilty person.
69	Nyili Uyu	Spirits living beneath the earth that bring natural death.
70	Nyobo-Somen	An evil spirit of other clans that hunts the soul (<i>yall</i>) of the deceased like a fox.
71	Orom	An evil spirit; the spirit of the dead.
72	Oram-Gengne Uyu	The spirit of one who has died, or is about to die, that disturbs the living.
73	Opo Poi	A spirit that watches those who break the taboos (<i>arii riinum</i>) of an observed ritual.
74	Opo-Uyu	Spirits.
75	Pagia Wiyu	A spirit that helps to foretell a person's future. In its ritual a cock is hung from the topmost post of the house, and people shoot at it with bows and arrows while the priest chants the names of callings (hunter, officer, orator...). Hitting the target during a calling foretells that future.
76	Pakki (Kiri-Kiate)	God of strength and courage.
77	Palo-Malo	Deity of strength.
78	Pane-Mane	Deity of wealth.
79	Pokan Dodu	An important god of the same rank as Dori.
80	Polo	The moon: a symbol of the beauty of the night and of goodness.
81	Pootee Hilo	Deity of prosperity.
82	Poomin Wiyu	The deity who accepts the sacrifice of small animals, such as pigs and goats, for well-being.
83	Poote Wiyu	The deity who accepts the sacrifice of large animals, such as cows (<i>seei</i>) and mithun (<i>sebe</i>), for well-being.
84	Potar Wiyu	The spirit to whom rituals are performed for protection and safety outside the home.
85	Pirya Uyu	The spirit that controls a person's mind and understanding according to its will.
86	Potor Wiyu	The spirit, and ritual, that prevents epidemics in the community and the village.
87	Pugo Polo	The moon.
88	Hinggo Polo	The moon.
89	Pui Donyi	The almighty sun.
90	Vji Donyi	The almighty sun.
91	Puri (Jigri-Riary)	God and goddess of luck and power.
92	Puyi	God of dawn.
93	Hinji	God of dawn.
94	Ramlo	The rituals and ceremonies for human well-being.

#	Name	Description
95	Ramne Wiyu	Ancestral spirits that come to a person with wishes or to bless. Its ritual requires different kinds of animals.
96	Reishi Wiyu	A spirit that brings illness again if the taboos were not observed properly.
97	Riaram Wiyu	The lord who does evil in society, especially by leading young people into casual relationships.
98	Rinte-Parte	A holy spirit that brings bumper harvests and increases the store of food.
99	Riumchingjering	The eldest son of Kulu-Kuryum, whose brothers were Riumching and Riumdo; a being of great wisdom and knowledge. As a mythological lord he divided himself into Seching Chingringchijing and Nyido Doringchyjing to look after his brothers: Riumching, who became Seching (the earth), and Riumdo, who became the peaceful Nyido-Kolo (the sky).
100	Ringte Uyu	Evil spirits, provoked by wrongdoing within a related family, that bring illness and bad luck to a person.
101	Ryalo Wiyu	The lord of spiritual blessing for the whole family.
102	Ryabu	The male form of an angel, from whom the Achi Ryabu Ponung troupe of men descends.
103	Sangru Wiyu	The spirit that can pardon past wrongdoing. In its ritual, with the help of a priest, a white or black hen is tied to a Tani or Tago tree; when the chanting is complete, the tree is cut down.
104	Seching Uyu	The earthly lord.
105	Svchii	The earthly lord.
106	Seh Uyu	The king or lord of the jungle.
107	Sebe Uyu	The lord of the mithun (<i>sebe</i> , <i>Bos frontalis</i>).
108	Seki Sebe Wiyu	The lord to whom a mithun is sacrificed for the blessing of the family.
109	Si Chenga	A male deity dwelling inside the earth.
110	Si-Donyi	The Earth and the Sun as the combined almighty. The Si-Donyi festival honours them to receive their blessing.
111	Sili	A spirit behind modern troubles, to whom money is offered at the altar in place of an egg or a chicken.
112	Si-Moro	The judge created by Doring Chijing (Dorichiji) as head of judicial authority, to preside whenever justice is needed.
113	Sichi-Nyodo Wiyu	The god of soil and rain. Siching and Nyodo, the earth and the sky, are the children of Kurium Kulu.
114	Soniyu Iyu	The spirit that leads to incest between sister and brother, or to unnatural death.
115	Sotung-Eraft	Spirits invoked to punish an enemy; also the name of this ritual.
116	Sotung Wiyu	A spirit that listens to the priest's chanting to bring bad luck when this is demanded. A wrongdoer may even die when this ritual is performed.
117	Taling Chane Wiyu	The spirit of the sky. (<i>Wiyu</i> are the spirits of unnatural death.)

#	Name	Description
118	Tamhilo	God of the fate and destiny of humankind.
119	Tamu Wiyu	A spirit of fertility that helps a family to grow in number.
120	Tanium Wiyu	A spirit of jealousy and ill will towards others. Rituals and sacrifices are made to turn it favourable.
121	Tangji Wiyu	The spirit, and ritual, that supports women's health.
122	Tapar Wiyu	A spirit that grants long life. In its ritual a hen is beheaded; if its body keeps moving for a long time, this foretells a long life.
123	Tapu Wiyu	A spirit that causes hearing problems, and whose blessing can cure them.
124	Tarbu Debu	The goddess of the household.
125	Uyu (Wiyu)	Spirit in general, good or bad.
126	Wiyu Mirang	A spirit that takes human form, in any shape it wishes.
127	Yachung Wiyu	Spirits that bring uncontrolled desire and the harm that follows from it.
128	Yarne Uyu	The deity who endows people with fame and abilities.
129	Yapom Wiyu	The spirit of high mountains and deep forests, which claims the forests, their animals and birds, and the rugged land as its own. It can carry a person away to places of its choosing, and its blessing can make a person a priest (nyibu).
130	Yalu Nanam	The ritual in which the priest brings back the soul of a living person from the realm of evil spirits, where some spirits hide human souls.
131	Yallung Uyu	Spirits that make people quarrelsome.
132	Yirne Wiyu	The spirit that encourages and increases the strength of a person and of the priest.
133	Yubu Wiyu	A spirit that blesses a person in every way. It has two forms: Yubu Botar, the lesser, and Yubu Bote, the greater.
134	Yullo Uyu	The great ceremony of peace, which invites happiness and prosperity.
135	Funying Uyu	A kind of spirit that easily causes harm in many ways.

D.2 Everyday and ritual register

Ritual-register words occur in chants and priestly speech. Tag them with the same POS as their everyday equivalent; in a ritual context their entity label is **CULT**.

English	Everyday word	Ritual word
sun	donyi	vjwng (vjww)
water, sea	isi	hai
pig	irik (vrik)	tachwng
poultry	porok	tabi
mithun	subu	svnyi

English	Everyday word	Ritual word
woman	nyimv	bvnyi
fish	ngui	ringchi
rat	kubung (kubuu)	bungpi
bird	pvtā	taachvk

D.3 Place names of Arunachal Pradesh

Districts, circles, towns and villages, in lowercase standard form. Every name is **PROPN** and a **LOC** span; multi-word names are one span.

aalo, anelih, anini, anjaw, arzoo, bagra, balemu, balijan, bameng, bana, bandardewa, baririjo, basar, bhalukpong, bilat, boleng, bomdila, bordumsa, borduria, buragaon, chaglagam, chambang, changlang, chetam, chongkham, dadam, dambuen, damin, daporijo, darak, daring, debeyer, deed, deomali, desali, dibang valley, digbak, dirang, dissing paso, diyun, dudunkhar, dumporijo, east kameng, east siang, etalin, gangte, geku, gelling, gensi, gepen, giba, goiliang, gumto, gusar, gyawepurang, hawai, hayuliang, hunli, itanagar, jairampur, jamin, jamiri, jang, jengging, kakoi, kalaktang, kamba, kamle, kamporijo, kangku, kantang, kanubari, kasim putu, katan, kaying, kharsang, khenewa, khimiyang, kibithoo, kimin, kodak, koloriang, kora, koronu, koyu, kra daadi, kurung kumey, lada, lathao, launu, lazū, lem̄mi, leparada, leporiang, likabali, limeking, liromoba, lohit, longding, longo, lower dibang valley, lower siang, lower subansiri, lumla, lyngok, mahadevpur, manchal, manmao, mariyang, maro, matengliang, mebo, mechuka, mengio, miao, mipi, monigong, mukto, nacho, nafra, nampong, namsai, namsing, namtok, nari, new seren, nilling, nyapin, old ziro, oyan, paglam, pakke-kessang, pakoti, palin, palling, pangin, pania, papum pare, parang, parbuk, paro, parsiparlo, pasighat, passa, payum, phassang, pidi, pijirang, pipsorang, pipu, piyong, pongchau, puchi geko, raga, renuk, richu krong, riga, rima-putok, roing, ruksin, rumgong, rupa, sagalee, sangram, sarli, seijosa, seppa, sewa, shergaon, shi-yomi, siang, silsango, singa, singchung, siyum, soha, sunpura, taksing, tali, taliha, tarasso, tato, tawang, tezu, thembang, thingbu, thrizino, tirap, tirbin, toru, tuting, upper siang, upper subansiri, vijoyanagar, wakka, wakro, walong, west kameng, west siang, yachuli, yagrung, yangte, yatdam, yazali, yingking, yomcha, yupia, zemithang, ziro.

D.4 Loanword transcription samples

These show how English words are spelled in Tagin (rules in §5.4).

English	Tagin (MRST)	English	Tagin (MRST)
Google	gugvl	Airbnb	eyvrbinbi
selfie	selpii	gadget	geyjet
meme	miim	freelancer	prilensvr
hashtag	hesteg	cryptocurrency	kriptokarvnsi
Zoom	juum	cloud computing	klaud kompiuting
podcast	podkest	Bluetooth	blu-tut
Wi-Fi	uai-pai	smartwatch	smartuach
emoji	imoji	e-wallet	i-ualet

D. Gazetteers

English	Tagin (MRST)	English	Tagin (MRST)
app	ep	self-driving	selp-draibing
metaverse	metabvrs	FaceTime	pestaim
blockchain	blockchein	deepfake	dippeke
chatbot	chetbot	NFT	nft
vlog	blog	algorithm	elgoridvm
Netflix	netfliks	quantum computing	kuantvm kompyuting
screenshot	skrinsot	hypothesis	haipotvsis
influencer	influensvr	derivative	deribetib
avatar	abatar	nanotechnology	nenoteknoloji
YouTube	yutyub	Big Bang	big beng
Instagram	instagram	machine learning	mvshin lvrning
reels	riils	artificial intelligence	artipisvl intelijens
TikTok	tiktok	DNA sequencing	dna sikuensing
Snapchat	snepchet	astrobiology	estrobaioloji
livestream	laibstrim	gravitational waves	grebitesvnel uebs
e-commerce	ikomvrs	gene editing	j in editin
drone	dron	neural network	nyurvvl netuork
smartphone	smartpon	statistical model	stetistikvl model
laptop	leptop	black hole	blek hol
dark matter	dark metvr	CRISPR	krispvr
string theory	istring tiori	zero gravity	jiro grabiti
Higgs boson	higs boson		

APPENDIX E

Kinship terms and clans

E.1 Kinship terms

All kinship terms are **NOUN** with entity label **O**.

Meaning	Standard	Dictionary	Meaning	Standard	Dictionary
father	abu	abu	mother	anv	ane
brother	achi	achi	sister	anyi / atv	anyi / ate
husband	nyilo	nyilu	wife	nying	nying
son	khu nyilo	khu nyilo	daughter	khu nyimv	khu nyime
grandfather, ancestor	atu, apa	atu / apa	grandmother	ayu	ayu
man	nyiga	nyiga	woman	nyimv	nyime
young man	yaapa	yaapa	girl	nyijar	nyijar
child, baby	anga	—	parents, ancestors	abu-apa	abuapa

The dictionary also lists *nikam* (old man) and *heu* (old woman). A kinship term that begins the name of a deity or ancestral being is part of that name: abu-tanyi is one token, **NOUN**, **CULT**.

E.2 Clans and lineages

Tagin clans and lineages include the following (lowercase annotation forms):

duchok, mosing-mosu, tamin, leyu, reri, natam-gyadu, nah, gyama, tache-tagia, tasi, dui, topo, tani-tator, cherom-chera, buning, heche, kodak-konia, pombu, bagang, aiyeng, paji, dulom, dugi.

Use	Tags
a clan as a group of people	PROPN, CULT
a clan name inside a person's name	PROPN, inside the PER span: takar/B-PER dugi/I-PER
a village named after a clan	PROPN, LOC

APPENDIX F

Worked examples

Each example is written token by token as *word/POS/NER*, followed by its English translation. The notes explain the decisions that are easy to get wrong.

F.1 The object marker for people and “person”

```
nyii/NOUN/O mwlww/PRON/O takar/PROPN/B-PER nyi/PART/O kaagia/VERB/O do/PART/O  
↪ ./PUNCT/O  
"Everybody likes Takar."
```

nyii is “people” (with mwlww, “everyone”). nyi marks Takar as the person who is liked (§11.1).

F.2 “Yes” and the object marker

```
vm/INTJ/O ,/PUNCT/O ngo/PRON/O ngo/PRON/O gv/PART/O napum/NOUN/O vm/PART/O  
↪ kaagia/VERB/O do/PART/O ./PUNCT/O  
"Yes, I like my nose."
```

The first vm is clause-initial and followed by a comma → INTJ “yes”. The second marks the object → PART. napum “nose” is an everyday body part → O (§14.5).

F.3 la as “and”

```
tara/PROPN/B-PER la/CCONJ/O tapa/PROPN/B-PER addin/NOUN/O baabam/VERB/O do/PART/O  
↪ ./PUNCT/O  
"Tara and Tapa are roasting meat together."
```

la joins two names → CCONJ, and each name is its own span (§15.4). addin “meat” being cooked is everyday food → O.

F.4 A full date

```
ngo/PRON/O 2024/NUM/B-DATE anying/NOUN/I-DATE gv/PART/O limi/NOUN/B-DATE  
↪ poolu/NOUN/I-DATE gv/PART/O  
12/NUM/B-DATE tarwk/NOUN/I-DATE alu/NOUN/I-DATE lo/PART/O vvkar/VERB/O re/PART/O  
↪ ./PUNCT/O  
"I will go out on the 12th of the month Limi, 2024."
```

There are three DATE spans; the particles gv and lo between them are O (§14.4). limi is a regional form of the February month name lwmi and is kept as written (Appendix C).

F.5 Clock time and acham “ten”

```
bww/PRON/0 ayu/NOUN/B-DATE acham/NUM/I-DATE apo/NOUN/I-DATE lo/PART/0 naam/NOUN/0
↪ aapa/VERB/0 ./PUNCT/0
"He came home at ten."
```

F.6 The literal sun

```
donyi/NOUN/0 v/PART/0 poklin/VERB/0 puku/PART/0 ./PUNCT/0
"The sun has risen."
```

The sun as a heavenly body → O. Only the deity is CULT (§11.2).

F.7 A person and an organisation

```
takar/PROPN/B-PER rajiv/PROPN/B-ORG gandhi/PROPN/I-ORG university/PROPN/I-ORG
↪ lo/PART/0 puri/VERB/0 do/PART/0 ./PUNCT/0
"Takar studies at Rajiv Gandhi University."
```

takar and the university are adjacent entities of different types, so each begins with B-. The person's name inside the university's name is not marked separately (§15.1).

F.8 A religious name

```
si-donyi/NOUN/B-CULT hilo/NOUN/I-CULT uyu/NOUN/I-CULT hv/PART/0 ...
```

Names of deities and spirits are NOUN with the label CULT (§10.3); the topic particle after the name is O.

F.9 nyii in a role noun

```
v/PART/0 ngui/NOUN/0 pyoknv/VERB/0 nyii/NOUN/0 rido/VERB/0 ./PUNCT/0
"He is a fish dealer."
```

pyoknv nyii “a person who sells” → nyii NOUN. Fish sold at the market is everyday food → O.

F.10 A verb and a noun chosen by the translation

```
ngom/PRON/0 achuk/NOUN/0 go/DET/0 tamsar/NOUN/0 jila/VERB/0 kv/PART/0 ./PUNCT/0
"Give me some advice."
```

tamsar is “advice” in the translation → NOUN, not VERB. jila “give” is a verb after another word → VERB (§10.2).

F.11 A time word with a loose translation

```
ngo/PRON/0 seeyu/NOUN/B-DATE iji/NOUN/0 hamma/VERB/0 la/PART/0 ./PUNCT/0
"I was stitching cloth in a dream [last night]."
```

seyyu means “last night” and is a time word → B-DATE, even though the translation renders the sentence as a dream (§11.2). iji “clothes” is an ordinary object → O. la after a verb → PART.

APPENDIX G

Decision register

Every question that was once open, or on which annotators once disagreed, is recorded here with its ruling. The rulings are part of the standard. New rulings are added under the procedure of Chapter 22.

ID	Question	Ruling	See
R1	What does <i>nyi</i> mark?	The object or recipient when it is a person ; never the agent. Always PART. “Person, people” is <i>nyii</i> , NOUN.	§3.2
R2	How is <i>nyi</i> read in chants without a translation?	Where it stands in a noun position (subject, before a determiner, before <i>vdw</i>), it is “people”: <i>nyii</i> , NOUN.	§11.1
R3	Is <i>nam</i> “you”?	Yes, in the sense “(to) you” → PRON. After a verb it is the verbal-noun ending → PART. Written for “house” it is corrected to <i>naam</i> .	§3.3
R4	Is <i>nga</i> an article “the”?	No. After a noun it is a pointing determiner (DET). Meaning “me, my” it is PRON.	§3.5
R5	Are everyday animals, foods and body parts CULT?	No: they are O. They are CULT only in a ritual, sacrificial or symbolic context.	§14.5
R6	Is standalone <i>donyi</i> the sun or the deity?	The sun, O, unless the sentence addresses or worships it as a deity, or it is part of a divine name (<i>donyi uyu</i> , <i>donyi-polo</i>). The same holds for <i>poolu</i> / <i>poolo</i> .	§14.5
R7	Is <i>seeyu</i> a time word?	Yes. It means “last night”: NOUN, B-DATE, also where a translation renders the sentence loosely (“in a dream”).	§11.2
R8	When may PROPON have entity label O?	Only for language names and the names of foreign peoples and groups. Paired chant words and ritual compounds that do not name something are tagged by meaning.	§10.4
R9	How are month names standardised?	The reference list gives the standard form; regional forms (<i>limi</i> , <i>lwchar</i> , <i>hiil</i> , <i>svvt</i> , <i>pwra</i> , <i>ralv</i> , and others) are kept as dialect variants with the same tags.	Appendix C
R10	What POS do names of deities, spirits and festivals take?	NOUN, with entity label CULT. Names of tribes, communities, languages and religions are PROPON.	§10.3

ID	Question	Ruling	See
R11	How is a homograph such as <i>siliguri</i> or <i>dugi</i> tagged?	By what it names in the passage: a being in a chant → CULT; a town or village → LOC; part of a person's name → PER; a clan → CULT.	§11.2
R12	Is <i>bolo</i> a determiner?	Yes, in every use, including “if” after a verb, like <i>olo</i> , <i>tolo</i> and <i>boolo</i> . One tag keeps the form consistent.	§10.5
R13	May template or synthetic sentences be used?	Never with invented names. With real entities, only if labelled synthetic in the data statement.	Ch. 8
R14	Are <i>kula</i> , <i>jila</i> , <i>dubv</i> , <i>hvla</i> particles?	No: they are VERB. <i>vla</i> is written <i>hvla</i> .	§10.2
R15	What class are <i>okv</i> , <i>hvbv</i> , <i>hvka</i> , <i>jeka</i> , <i>nyila</i> ?	Always CCONJ. <i>la</i> is CCONJ only between two nouns, names or pronouns, otherwise PART.	§10.6
R16	How is the plural marker written?	<i>vdw</i> , always its own token, PART, O. <i>adw</i> “short” is written <i>adww</i> .	§6.1
R17	What is <i>vm</i> / <i>um</i> ?	<i>um</i> inside a phrase is written <i>vm</i> . <i>vm</i> is PART (object marker, “that”), or INTJ “yes” when clause-initial before a comma.	§11.2
R18	Are capital letters used?	No. All text is lowercase; names are identified by their tags.	§4.1
R19	How are scripture references tokenized?	One NUM token per reference; a range is NUM + dash + NUM.	§7.1
R20	What is <i>acham</i> ?	“group” → NOUN; “ten” → NUM; “sleep” → VERB.	§11.2
R21	Is a title before a name part of the name?	Yes, when the two together are how the person is known; otherwise no.	§15.3
R22	How are <i>he lo</i> / <i>heloo</i> and <i>aalle</i> / <i>alle</i> written and tagged?	<i>heloo</i> and <i>alle</i> : INTJ, O.	§6.1
R23	Are clothes (<i>iji</i>) and palm leaf (<i>tawk</i>) cultural artifacts?	No: ordinary objects, NOUN, O.	§14.5
R24	How are <i>jius</i> and <i>jentail</i> tagged?	PROPN, O (names of foreign groups).	§10.4
R25	Is <i>poolu alu</i> one token?	Yes: <i>poolu-alu</i> , the name of a day, B-DATE.	§14.4
R26	What is <i>kam</i> ?	After a verb → VERB. Linking two clauses (“if”) → CCONJ.	§11.2
R27	What is <i>aku</i> ?	ADJ (“one, some”), not PRON or DET.	§10.5
R28	Are <i>so</i> , <i>ho</i> , <i>hoka</i> , <i>soka</i> case particles?	No: <i>so</i> , <i>ho</i> , <i>hoka</i> are DET; <i>soka</i> is ADV.	§3.2

ID	Question	Ruling	See
R29	How many vowels does Tagin have?	Seven: a, e, i, o, u, v [ə] and w [i], each short or long. e [e] and v [ə] are different vowels, although the dictionary writes both as <i>e</i> .	§2.1
R30	How many hyphens may a token contain?	A native Tagin word : at most one. A hyphen joins exactly two parts; further parts are separate words: rokpu-podo polo, not rokpu-podo-polo. Foreign words may contain more than one (R31).	§4.3
R31	How are foreign words with several parts written?	Hyphenated foreign names and terms are one token and keep their hyphens: majar-i-sarip, pruf-op-uvrk. Abbreviations keep their original letters , in lowercase and without hyphens: nft, upi, dna, cd; never letter-name spellings such as en-ep-ti or si-di.	§4.3
R32	Is a double hyphen -- inside a word ever correct?	No. It is a typo and is written as one hyphen: boosik--bomik → boosik-bomik.	§4.3
R33	How is “you all” written?	nunu, also at the start of longer words (nunua, nununo, nununogv). nonu is no longer a standard form.	§3.3
R34	Is “you two” nonyi or nunyi?	nunyi. This settles the conflict between the Mooya conversion table and the pronoun list of edition 3.	§3.3
R35	Is a name joined to nyi one token?	No. A name and nyi are always two tokens, for people and for God alike: jisu nyi, pwknvyarnv nyi, paul nyi, kristo nyi. jisunyi and pwknvyarnvnnyi are no longer lexicalised exceptions.	§7.2
R36	How is “oneself” written?	atu boongv, in line with bongv → boongv in the normalization table.	§6.1
R37	Are the future re and the command to written apart from the verb?	Yes: vngjire → vngji re, vngroto → vngro to. In <i>-roto</i> , ro stays with the verb and to is split; a <i>roto</i> written apart after a verb joins that verb (vngyu roto → vngyuro to). Exceptions: jire standing alone stays one word, and adu roto is written as it is.	§7.2
R38	How is <i>koching</i> written?	kooching, also inside longer words (koochinglo, koochingso); <i>hokochingbv</i> is ho kooching bv.	§5.2
R39	How is Mooya <i>oguaingbv</i> written?	hugu aing bv in every spelling (<i>oguaingbv</i> , <i>nonu-oguaingbv</i>); <i>oguaingnv</i> is hugu aing nv.	§5.2

ID	Question	Ruling	See
R40	What does the Mooya–Mookwng word list decide?	It gives the Mookwng spelling of Mooya words and is used for conversion, reading capitalised sentence-initial forms separately (<i>Vdwlo</i> “when” → <i>hwdwlo</i>). Where it contradicts this book, the book wins: <i>okv</i> stays (R15), the lowercase plural <i>vdw</i> stays (R16), and <i>um</i> becomes <i>vm</i> (R17).	§5.2
R41	Are <i>lvgabv</i> , <i>bunugv</i> , <i>hugulo</i> , <i>gvri la</i> and <i>ngogv</i> single words?	No. The particle is written apart: <i>lvga bv</i> “because, for” (<i>hugu lvga bv</i> , <i>hvkv lvga bv</i>), <i>bunu gv</i> “their”, <i>hugu lo</i> “where”, <i>gvri la</i> , <i>ngo gv</i> “my”. They are removed from the lists of single words (Appendix A, §7.2).	§7.2
R42	Which forms that look like a word and a particle stay whole?	The words listed in §7.2 (<i>hvla</i> , <i>hvkv</i> , <i>vdw</i> , <i>sinv</i> , <i>alvbv</i> , <i>atugv</i> ...). Other pairs are different words when joined and when apart, and both stay as written (<i>soka</i> / <i>so ka</i> , <i>nyilo</i> / <i>nyi lo</i>). Everything else follows the general rule: split.	§7.2
R43	Are paired compounds from scripture hyphenated?	Yes, like other paired compounds, with one hyphen: <i>ajin–anying</i> , <i>rinam–daknam</i> , <i>ridung–singdung</i> , <i>mdur–ridur</i> , <i>dvnam–tvngnam</i> , <i>loonyi–loum</i> .	§4.3
R44	How are units of measure and English notes in translated text written?	Units are spelled as Tagin speakers say them: <i>kilomitar</i> , <i>mitar</i> , <i>pit</i> (feet), <i>mails</i> ; numbers are written in Tagin (<i>hejar gola angyi kilomitar</i>). English notes left in a Tagin text are removed or rewritten in Tagin.	§5.4

APPENDIX H

Quick reference card

Topic	Rule
Letters	a b c h d e g h i j k l m n n g ny o p r s t u v w y; seven vowels a e i o u v w, with v = ə, w = i; long vowels doubled; tone not written
Case	everything lowercase; straight quotes only
Hyphen	joins two parts of one lexical unit → one token (si-donyi, poolu-a lu); never more than one hyphen in a Tagin word (rokpu-podo polo); foreign names and terms keep all their hyphens (majar-i-sarip); abbreviations keep their original letters (nft, upi); -- is a typo
Tokens	split at spaces; every punctuation mark is a token; particles are separate tokens (vdw gv, not vdwgv; jisv nyi; vngji re; vngro to; lvga bv; bunu gv); words kept whole and joined/apart pairs in §7.2
Sentences	one utterance each; always end in punctuation; chants split into verse lines
POS order	PUNCT → X → NUM → INTJ → PART → CCONJ → PRON → DET → deity/festival name = NOUN → other name = PROPN → VERB → ADJ / ADV → NOUN
Never	ADP, AUX, SCONJ, SYM
Meaning decides	tag by the translation; if uncertain, mark for adjudication
NER types	PER, LOC, ORG, DATE, CULT; BIO labels; flat spans; longest name wins
Always O	particles, pronouns, conjunctions, determiners, punctuation, interjections
CULT	deities, spirits, rituals, festivals, traditional roles, tribes, clans, cultural artifacts; animals and food <i>only</i> in ritual context

Word	Reading → tag
nyi / nyii	marker for a person → PART / “person” → NOUN
nam	after a verb → PART; “you” → PRON; “house” → spell naam
nga	after a noun, pointing → DET; “me, my” → PRON
vm	object marker, “that” → PART; clause-initial “yes” before a comma → INTJ
la	between two nouns / names / pronouns → CCONJ; otherwise PART
acham	group → NOUN; ten → NUM; sleep → VERB
bolo	DET in every use, including “if”
so, ho / soka	DET / ADV (not case particles)
donyi, poolu	sun, moon → NOUN, O; in a divine name → CULT; month → DATE
kula, jila, dubv, hvla	VERB
okv, hvbv, hvka, jeka, nyila	CCONJ

Word	Reading → tag
vdw	plural → PART, O, always its own token

APPENDIX I

Glossary of terms

Term	Meaning
Adjudication	Resolving disagreements between annotators by applying this book, and recording the reason.
BIO	The labelling scheme that marks the Beginning and Inside of each entity, and tokens Outside any entity.
Bitext	Tagin sentences paired with their translations.
CARE	Principles for Indigenous data governance: Collective benefit, Authority to control, Responsibility, Ethics.
Classifier	A prefix on a numeral showing the shape or kind of thing counted.
CoNLL-U	The tab-separated file format of the Universal Dependencies project.
CULT	The entity type for culturally significant beings, practices, roles, groups and artifacts.
FAIR	Principles for research data: Findable, Accessible, Interoperable, Reusable.
Gold data	Data annotated independently by two people and adjudicated.
κ (kappa)	Cohen’s measure of agreement between two annotators, corrected for chance.
Lexicalised	Fixed as a single word in the language, even if it was once two.
MRST	Modified Roman Script for Tagin, the standard spelling (also called the GinLish spelling).
Mooya / Mookwng	The two spelling traditions of the Tagin Bible translations.
NFC	Unicode Normalization Form C, the standard way of storing accented and combined characters.
Particle	A short grammatical word that follows another word (case, tense, negation...); tagged PART.
Register	A style of language for a setting, e.g. everyday or ritual.
Silver data	Data annotated by one person, or by a model and checked by one person.
Span	One or more consecutive tokens forming one entity.
Token	One unit of text that receives one POS tag and one entity label.
UPOS	The Universal Dependencies part-of-speech tagset.

APPENDIX J

Tagin and Galo correspondences

Tagin and Galo are closely related Tani languages with the same set of sounds, written in different Roman scripts. This appendix supports parallel and comparative work. It is not a conversion table: Galo text is never converted into Tagin (§5.5).

Sound	Tagin	Galo	Example (Tagin / Galo)
ŋ	ng	q	ngo / qo “I”
ɲ	ny	x	anyi / axi
ə	v	v	vm / `vm
i	w	w	bww / `bww
long vowel	doubled	doubled	naam
double consonant	not written	written	okv / okkv
tone	not written	leading backtick `	la / `la

Shared grammatical words:

- **Pronouns:** ngo / qo, no / no, bww / `bww, ngunu / `qunu, bulu / `bulu
- **Particles:** lo, vm, gv, bv, ku, to, ma / maa, re / `ree, do / `duu, la / `la
- **Determiner:** go
- **Conjunctions:** okv / okkv, go!a, gv!a

Galo annotated under this book follows the same POS and NER rules, with these Galo-specific conventions:

- Galo acham + a number word is one hyphenated NUM token: acham-`kanv (70), acham-`piinv (80), acham-`keqqaa (90).
- `!aa is written `!a.
- The name of the ancestor Abotani is one token, abo-tanii (NOUN, B-CULT), as Tagin abu-tanyi.
- A title before a name forms one PER span: axi aqam = B-PER I-PER.

APPENDIX K

Sources

Tagin sources

- *Tagin Dictionary*, with grammar chapters: Alphabets and Phonology; Morphology and Syntax; Noun; Pronoun; Verb and Tense; Active and Passive Voice; Adverb; Article; Gerund; Conjunction; Mood; Adjectives; The Countings.
- *Tagin Glossary*.
- *Tagin Gaam Kitab*, school readers published by the State Council of Educational Research and Training (SCERT), Arunachal Pradesh.
- The Tagin Bible translations in the Mooya and Mookwng spelling traditions.
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Reference facts

Classification, codes, speaker numbers and status: Census of India 2011; Glottolog tagi1241; Ethnologue tgj; UNESCO Atlas of the World's Languages in Danger.

APPENDIX L

Edition history

Edition	Date	Changes
5.0	2026-09-29	Completion of the native speaker’s review of the Tagin New Testament. Particles are written apart in forms earlier listed as single words: lvga bv, bunu gv, hugu lo, gvri la, ngo gv; lvgabv, bunugv, hugulo, gvri la and ngogv are removed from Appendix A and from the lexicalised exceptions, and the book’s examples are updated. New lists of words kept whole and of pairs that are different words joined and apart. New examples of paired compounds, Mooya conversions (mvnglwk → mwnglwk, kokwng → kookwng, logo → loogo) and loanwords for units of measure (kilomitar, mitar, pit, mails). Normalization table, validation checks and quick reference updated. Major edition because earlier rules are reversed. Decision register R41–R44.
4.0	2026-09-29	Rulings from a native speaker’s review of the Tagin New Testament (Mooya spelling). “You all” is nunu (also at the start of longer words) and “you two” is nunyi; nonu is removed from the pronoun lists. A name and nyi are two tokens (jisu nyi, pwknvyarnv nyi); jisunyi and pwknvyarnvnnyi are no longer lexicalised exceptions. The future re and the command to are written apart from the verb (vngji re, vngro to), with the exceptions jire and adu roto. “Oneself” is atu boongv, and the passive marker is written boongv. Mooya conversions added: koching → kooching, oguaingbv → hugu aing bv, nonu → nunu; the Mooya–Mookwng word list and its limits described. Validation checks and quick reference updated. Major edition because earlier rules are reversed. Decision register R33–R40.
3.1	2026-09-28	Hyphen rule refined: the one-hyphen limit applies to native Tagin words ; hyphenated foreign names and terms keep all their hyphens (majar-i-sarip, pruf-op-uvrk). Abbreviations keep their original letters in lowercase (nft, upi, cd) instead of letter-name spellings (en-ep-ti, yu-pi-ai, si-di / sidi). A double hyphen -- is a typo and is written as one (boosik--bomik → boosik-bomik). Decision register R30–R32.
3.0	2026-09-28	General-purpose standard. Tagin described with seven vowels (a e i o u v w); e added to the standard letters. A token contains at most one hyphen (rokpu-podo polo), except abbreviations written with hyphenated English letter names (en-ep-ti). All open questions decided and recorded in the decision register (Appendix G): everyday animals, food and body parts are O; standalone sun and moon are O; seeyu is the time word “last night”; untranslated nyi in noun positions is nyii; PROPN with O limited to language and foreign-group names; month names standardised with regional variants; homographs tagged by context; bo lo is DET in every use; kam after a verb is VERB; aku is ADJ; so, ho, hoka are DET and soka ADV; names of deities and festivals are NOUN + CULT. New chapters: difficult entity cases, annotators and adjudication, agreement measurement, model-assisted annotation, validation, maintenance of the standard, access levels and take-down. New formats: CoNLL-U. New appendices: decision register, quick reference card. Deity descriptions edited for clarity; place list cleaned and sorted.
2.1	2026-09-28	nam “(to) you” is PRON; nam for “house” is corrected to naam; nga is a pointing determiner (DET), not an article, and “me, my” is PRON; kunyibokar is written kunyi-bokar.

Edition	Date	Changes
2.0	2026-09-28	Full rewrite: language profile; sound system and dialect alternations; grammar sketch with case, pronoun and tense-aspect-mood tables; conversion rules for the dictionary, Mooya, IPA and loanword spellings; POS and NER decision procedures; mapping from older tagsets; crosswalk to a fine-grained NER scheme; numerals and classifiers; calendar; gazetteers; kinship and clans; worked examples; workflow, quality targets, data statement; stewardship and ethics. <i>nyi</i> established as the object / recipient marker for people.
1.x	2026-09-27	First edition: spelling normalization (<i>vh</i> → <i>v</i> , <i>po lo</i> → <i>poolo</i> , <i>tani</i> → <i>tanyi</i> , <i>adw</i> → <i>vdw</i> / <i>adww</i> , <i>nyi</i> → <i>nyii...</i>), POS and NER conventions, chant splitting, lowercase text.